

6-8-9

1. If $\int_0^k \frac{x}{x^2 + 4} dx = \frac{1}{2} \ln 4$, where $k > 0$, then $k =$

(A) 0

(B) $\sqrt{2}$

(C) 2

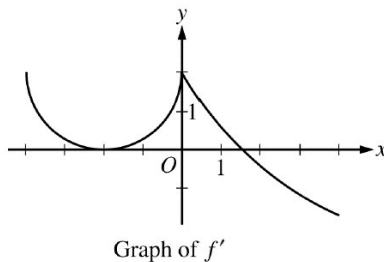
(D) $\sqrt{12}$



(E) $\frac{1}{2} \tan(\ln \sqrt{2})$

6-8-9

2.



The derivative of a function f is defined by $f'(x) = \begin{cases} g(x) & \text{for } -4 \leq x \leq 0 \\ 5e^{-x/3} - 3 & \text{for } 0 < x \leq 4 \end{cases}$.

The graph of the continuous function f' , shown in the figure above, has x -intercepts at $x = -2$ and $x = 3 \ln\left(\frac{5}{3}\right)$. The graph of g on $-4 \leq x \leq 0$ is a semicircle, and $f(0) = 5$.

- (a) For $-4 < x < 4$, find all values of x at which the graph of f has a point of inflection. Justify your answer.
- (b) Find $f(-4)$ and $f(4)$.
- (c) For $-4 \leq x \leq 4$, find the value of x at which f has an absolute maximum. Justify your answer.

Part A

1 point is earned if identifies $x = -2$ or $x = 0$

1 point is earned for answer with justification

f' changes from decreasing to increasing at $x = -2$ and from increasing to decreasing at $x = 0$. Therefore, the graph of f has points of inflections at $x = -2$ and $x = 0$.

✓

0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned if identifies $x = -2$ or $x = 0$

1 point is earned for answer with justification

f' changes from decreasing to increasing at $x = -2$ and from increasing to decreasing at $x = 0$. Therefore, the graph of f has points of inflections at $x = -2$ and $x = 0$.

Part B

1 point is earned for integral of $f(-4)$

1 point is earned for value of $f(-4)$

1 point is earned for integral of $f(4)$

1 point is earned for antiderivative of $f(4)$

1 point is earned for value of $f(4)$

$$\begin{aligned} f(-4) &= 5 + \int_0^{-4} g(x) dx \\ &= 5 - (8 - 2\pi) = 2\pi - 3 \end{aligned}$$

6-8-9

$$\begin{aligned}
 f(4) &= 5 + \int_0^4 (5e^{-x/3} - 3)dx \\
 &= 5 + (-15e^{-x/3} - 3x) \Big|_{x=0}^{x=4} \\
 &= 8 - 15e^{-4/3}
 \end{aligned}$$

✓

0	1	2	3	4	5
---	---	---	---	---	---

The student response earns all of the following points:

1 point is earned for integral of $f(-4)$

1 point is earned for value of $f(-4)$

1 point is earned for integral of $f(4)$

1 point is earned for antiderivative of $f(4)$

1 point is earned for value of $f(4)$

$$\begin{aligned}
 f(-4) &= 5 + \int_0^{-4} g(x)dx \\
 &= 5 - (8 - 2\pi) = 2\pi - 3
 \end{aligned}$$

$$\begin{aligned}
 f(4) &= 5 + \int_0^4 (5e^{-x/3} - 3)dx \\
 &= 5 + (-15e^{-x/3} - 3x) \Big|_{x=0}^{x=4} \\
 &= 8 - 15e^{-4/3}
 \end{aligned}$$

Part C

1 point is earned for the answer

1 point is earned for the justification

Since $f'(x) > 0$ on the intervals $-4 < x < -2$ and $-2 < x < 3 \ln\left(\frac{5}{3}\right)$, f is increasing on the interval $-4 \leq x \leq 3 \ln\left(\frac{5}{3}\right)$.

Since $f'(x) < 0$ on the interval $3 \ln\left(\frac{5}{3}\right) < x < 4$, f is decreasing on the interval $3 \ln\left(\frac{5}{3}\right) \leq x \leq 4$.

Therefore, f has an absolute maximum at $x = 3 \ln\left(\frac{5}{3}\right)$.

✓

0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the answer

1 point is earned for the justification

6-8-9

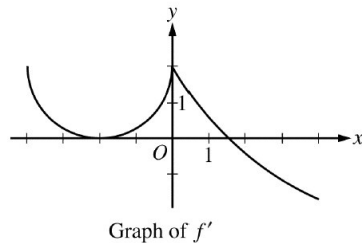
Since $f'(x) > 0$ on the intervals $-4 < x < -2$ and $-2 < x < 3 \ln\left(\frac{5}{3}\right)$, f is increasing on the interval $-4 \leq x \leq 3 \ln\left(\frac{5}{3}\right)$.

Since $f'(x) < 0$ on the interval $3 \ln\left(\frac{5}{3}\right) < x < 4$, f is decreasing on the interval $3 \ln\left(\frac{5}{3}\right) \leq x \leq 4$.

Therefore, f has an absolute maximum at $x = 3 \ln\left(\frac{5}{3}\right)$.

6-8-9

3.



The derivative of a function f is defined by $f'(x) = \begin{cases} g(x) & \text{for } -4 \leq x \leq 0 \\ 5e^{-x/3} - 3 & \text{for } 0 < x \leq 4 \end{cases}$.

The graph of the continuous function f' , shown in the figure above, has x -intercepts at $x = -2$ and $x = 3 \ln\left(\frac{5}{3}\right)$. The graph of g on $-4 \leq x \leq 0$ is a semicircle, and $f(0) = 5$.

Find $f(-4)$ and $f(4)$.

Part B

1 point is earned for integral of $f(-4)$

1 point is earned for value of $f(-4)$

1 point is earned for integral of $f(4)$

1 point is earned for antiderivative of $f(4)$

1 point is earned for value of $f(4)$

$$\begin{aligned} f(-4) &= 5 + \int_0^{-4} g(x) dx \\ &= 5 - (8 - 2\pi) = 2\pi - 3 \end{aligned}$$

$$\begin{aligned} f(4) &= 5 + \int_0^4 (5e^{-x/3} - 3) dx \\ &= 5 + (-15e^{-x/3} - 3x) \Big|_{x=0}^{x=4} \\ &= 8 - 15e^{-4/3} \end{aligned}$$



0	1	2	3	4	5
---	---	---	---	---	---

The student response earns five of the following points:

1 point is earned for integral of $f(-4)$

1 point is earned for value of $f(-4)$

1 point is earned for integral of $f(4)$

1 point is earned for antiderivative of $f(4)$

1 point is earned for value of $f(4)$

$$\begin{aligned} f(-4) &= 5 + \int_0^{-4} g(x) dx \\ &= 5 - (8 - 2\pi) = 2\pi - 3 \end{aligned}$$

6-8-9

$$\begin{aligned} f(4) &= 5 + \int_0^4 \left(5e^{\frac{-x}{3}} - 3 \right) dx \\ &= 5 + \left(-15e^{\frac{-x}{3}} - 3x \right) \Big|_{x=0}^{x=4} \\ &= 8 - 15e^{\frac{-4}{3}} \end{aligned}$$

6-8-9

4. Let f be a function that is differentiable throughout its domain and that has the following properties.

(i) $f(x+y) = \frac{f(x)+f(y)}{1-f(x)f(y)}$ for all real numbers x , y , and $x+y$ in the domain of f

(ii) $\lim_{h \rightarrow 0} f(h) = 0$

(iii) $\lim_{h \rightarrow 0} \frac{f(h)}{h} = 1$

(a) Show that $f(0) = 0$.

(b) Use the definition of the derivative to show that $f'(x) = 1 + [f(x)]^2$. Indicate clearly where properties (i), (ii) and (iii) are used.

(c) Find $f(x)$ by solving the differential equation in part (b).

Part A

1 point is earned for to show $f(0) = 0$

Method 1:

f is continuous at 0, so $f(0) = \lim_{x \rightarrow 0} f(x) = 0$

Method 2:

$$f(0) = f(0+0) = \frac{f(0)+f(0)}{1-f(0)f(0)}$$

$$f(0) (1 - [f(0)]^2) = 2f(0)$$

$$f(0) (-1 - [f(0)]^2) = 0$$

$$f(0) = 0$$



0

1

The student response earns all of the following points:

1 point is earned for to show $f(0) = 0$

Method 1:

f is continuous at 0, so $f(0) = \lim_{x \rightarrow 0} f(x) = 0$

Method 2:

$$f(0) = f(0+0) = \frac{f(0)+f(0)}{1-f(0)f(0)}$$

$$f(0) (1 - [f(0)]^2) = 2f(0)$$

$$f(0) (-1 - [f(0)]^2) = 0$$

$$f(0) = 0$$

Part B

1 point is earned for definition of $f'(x)$

1 point is earned for clearly uses (i)

6-8-9

1 point is earned for clearly uses (ii)

1 point is earned for clearly uses (iii)

1 point is earned for algebraic manipulation to answer

$$\begin{aligned}
 f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\
 &= \lim_{h \rightarrow 0} \frac{\frac{f(x)+f(h)}{1-f(x)f(h)} - f(x)}{h} \quad [\text{By (i)}] \\
 &= \lim_{h \rightarrow 0} \left[\frac{f(h)}{h} \cdot \frac{1+[f(x)]^2}{1-f(x)f(h)} \right] \\
 &= 1 \cdot \frac{1+[f(x)]^2}{1-f(x) \cdot 0} \quad [\text{By (iii) \& (ii)}] \\
 &= 1 + [f(x)]^2
 \end{aligned}$$



0	1	2	3	4	5
---	---	---	---	---	---

The student response earns all of the following points:

1 point is earned for definition of $f'(x)$

1 point is earned for clearly uses (i)

1 point is earned for clearly uses (ii)

1 point is earned for clearly uses (iii)

1 point is earned for algebraic manipulation to answer

$$\begin{aligned}
 f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\
 &= \lim_{h \rightarrow 0} \frac{\frac{f(x)+f(h)}{1-f(x)f(h)} - f(x)}{h} \quad [\text{By (i)}] \\
 &= \lim_{h \rightarrow 0} \left[\frac{f(h)}{h} \cdot \frac{1+[f(x)]^2}{1-f(x)f(h)} \right] \\
 &= 1 \cdot \frac{1+[f(x)]^2}{1-f(x) \cdot 0} \quad [\text{By (iii) \& (ii)}] \\
 &= 1 + [f(x)]^2
 \end{aligned}$$

Part C

1 point is earned if separates variables

1 point is earned for integration, with C

1 point is earned for c evaluated

Method 1:

6-8-9

Let $y = f(x)$; $\frac{dy}{dx} = 1 + y^2$

$$\frac{dy}{1+y^2} = dx$$

$$\tan^{-1} y = x + C$$

$$y = \tan(x + C)$$

$$f(0) = 0 \Rightarrow C = 0 \text{ [or } C = n\pi, n \in \mathbb{Z}]$$

$$f(x) = \tan x \text{ [or } f(x) = \tan(x + n\pi)]$$

or

Method 2:

Guess that $f(x) = \tan x$

$$1 + [f(x)]^2 = 1 + \tan^2 x = \sec^2 x = f'(x)$$

$$f(0) = \tan(0) = 0$$

Since the solution to the D.E. is unique $f(x) = \tan x$ is the solution.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned if separates variables

1 point is earned for integration, with C

1 point is earned for c evaluated

Method 1:

Let $y = f(x)$; $\frac{dy}{dx} = 1 + y^2$

$$\frac{dy}{1+y^2} = dx$$

$$\tan^{-1} y = x + C$$

$$y = \tan(x + C)$$

$$f(0) = 0 \Rightarrow C = 0 \text{ [or } C = n\pi, n \in \mathbb{Z}]$$

$$f(x) = \tan x \text{ [or } f(x) = \tan(x + n\pi)]$$

or

Method 2:

Guess that $f(x) = \tan x$

$$1 + [f(x)]^2 = 1 + \tan^2 x = \sec^2 x = f'(x)$$

$$f(0) = \tan(0) = 0$$

Since the solution to the D.E. is unique $f(x) = \tan x$ is the solution.

6-8-9

5. Let f be a function that is differentiable throughout its domain and that has the following properties.

(i) $f(x+y) = \frac{f(x)+f(y)}{1-f(x)f(y)}$ for all real numbers x, y , and $x+y$ in the domain of f

(ii) $\lim_{h \rightarrow 0} f(h) = 0$

(iii) $\lim_{h \rightarrow 0} \frac{f(h)}{h} = 1$

Find $f(x)$ by solving the differential equation in part (b).

Part C

1 point is earned if separates variables

1 point is earned for integration, with C

1 point is earned for c evaluated

Method 1:

$$\text{Let } y = f(x); \frac{dy}{dx} = 1 + y^2$$

$$\frac{dy}{1+y^2} = dx$$

$$\tan^{-1} y = x + C$$

$$y = \tan(x + C)$$

$$f(0) = 0 \Rightarrow C = 0 \text{ [or } C = n\pi, n \in \mathbb{Z}]$$

$$f(x) = \tan x \text{ [or } f(x) = \tan(x + n\pi)]$$

Method 2:

Guess that

$$f(x) = \tan x$$

$$1 + [f(x)]^2 = 1 + \tan^2 x = \sec^2 x = f'(x)$$

$$f(0) = \tan(0) = 0$$

Since the solution to the D.E. is unique $f(x) = \tan x$ is the solution.



0	1	2	3
---	---	---	---

The student response earns three of the following points:

1 point is earned if separates variables

1 point is earned for integration, with C

1 point is earned for c evaluated

Method 1:

6-8-9

$$\text{Let } y = f(x); \frac{dy}{dx} = 1 + y^2$$

$$\frac{dy}{1+y^2} = dx$$

$$\tan^{-1} y = x + C$$

$$y = \tan(x + C)$$

$$f(0) = 0 \Rightarrow C = 0 \text{ [or } C = n\pi, n \in \mathbb{Z}]$$

$$f(x) = \tan x \text{ [or } f(x) = \tan(x + n\pi)]$$

Method 2:

Guess that

$$f(x) = \tan x$$

$$1 + [f(x)]^2 = 1 + \tan^2 x = \sec^2 x = f'(x)$$

$$f(0) = \tan(0) = 0$$

Since the solution to the D.E. is unique $f(x) = \tan x$ is the solution.

6. Consider the function f given by $f(x) = xe^{-x^2}$ for all real numbers x .
Find the antiderivative of f .

Part B

One point is earned for the antiderivative

$$\int xe^{-x^2} dx = -\frac{1}{2}e^{-x^2} + C$$



0	1
---	---

The student earns all of the following points:

One point is earned for the antiderivative

$$\int xe^{-x^2} dx = -\frac{1}{2}e^{-x^2} + C$$

7. $\int (e^x + e) dx =$

(A) $e^x + C$

(B) $2e^x + C$

(C) $e^x + e + C$

(D) $e^{x+1} + ex + C$

(E) $e^x + ex + C$



6-8-9

8. If the second derivative of f is given by $f''(x) = 2x - \cos x$, which of the following could be $f(x)$?

(A) $\frac{x^3}{3} + \cos x - x + 1$ ✓

(B) $\frac{x^3}{3} - \cos x - x + 1$

(C) $x^3 + \cos x - x + 1$

(D) $x^2 - \sin x + 1$

(E) $x^2 + \sin x + 1$

9. Let R be the region in the first quadrant under the graph of $y = \frac{x}{x^2+2}$ for $0 \leq x \leq \sqrt{6}$.

Find the area of R .

Part A

1 point is earned for correct integral

1 point is earned for antiderivative

1 point is earned for evaluation

$$\begin{aligned} A &= \int_0^{\sqrt{6}} \frac{x}{x^2+2} dx \\ &= \frac{1}{2} \ln(x^2+2) \Big|_0^{\sqrt{6}} \\ &= \frac{1}{2} \ln 8 - \frac{1}{2} \ln 2 = \ln 2 \end{aligned}$$

✓

0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for correct integral

1 point is earned for antiderivative

1 point is earned for evaluation

$$\begin{aligned} A &= \int_0^{\sqrt{6}} \frac{x}{x^2+2} dx \\ &= \frac{1}{2} \ln(x^2+2) \Big|_0^{\sqrt{6}} \\ &= \frac{1}{2} \ln 8 - \frac{1}{2} \ln 2 = \ln 2 \end{aligned}$$

10. $\int \sec x \tan x \, dx =$

(A) $\sec x + C$ ✓

(B) $\tan x + C$

(C) $\frac{\sec^2 x}{2} + C$

(D) $\frac{\tan^2 x}{2} + C$

(E) $\frac{\sec^2 x \tan^2 x}{2} + C$

6-8-9

A particle moves along the x -axis in such a way that its acceleration at time t for $t \geq 0$ is given by $a(t) = 4\cos(2t)$. At time $t = 0$, the velocity of the particle is $v(0) = 1$ and its position is $x(0) = 0$.

11. Write an equation for the position $x(t)$ of the particle.

Part B

1 point is earned for integrand

1 point is earned for $x(t) = -\cos 2t + t + C$ or $x(t) = -\cos 2t + t$

1 point is earned for answer

$$x(t) = \int 2 \sin 2t + 1 dt$$

$$x(t) = -\cos 2t + t + C$$

$$x(0) = 0 \Rightarrow C = 1$$

$$x(t) = -\cos 2t + t + 1$$



0	1	2	3
---	---	---	---

The student response earns three of the following points:

1 point is earned for integrand

1 point is earned for $x(t) = -\cos 2t + t + C$ or $x(t) = -\cos 2t + t$

1 point is earned for answer

$$x(t) = \int 2 \sin 2t + 1 dt$$

$$x(t) = -\cos 2t + t + C$$

$$x(0) = 0 \Rightarrow C = 1$$

$$x(t) = -\cos 2t + t + 1$$

6-8-9

12. Write an equation for the velocity $v(t)$ of the particle.
-

Part A

1 point is earned for integrand

1 point is earned for $v(t) = 2 \sin 2t + C$ or $v(t) = 2 \sin 2t$

1 point is earned for answer

$$v(t) = \int 4 \cos 2t \, dt$$

$$v(t) = 2 \sin 2t + C$$

$$v(0) = 1 \Rightarrow C = 1$$

$$v(t) = 2 \sin 2t + 1$$



0	1	2	3
---	---	---	---

The student response earns three of the following points:

1 point is earned for integrand

1 point is earned for $v(t) = 2 \sin 2t + C$ or $v(t) = 2 \sin 2t$

1 point is earned for answer

$$v(t) = \int 4 \cos 2t \, dt$$

$$v(t) = 2 \sin 2t + C$$

$$v(0) = 1 \Rightarrow C = 1$$

$$v(t) = 2 \sin 2t + 1$$

6-8-9

13. A particle moves along the x -axis so that its velocity at any time $t \geq 0$ is given by $v(t) = 1 - \sin(2\pi t)$. Find the position $x(t)$ of the particle at any time t if $x(0) = 0$.

Part C

2 points are earned for the correct antiderivative of $v(t)$

1 point is earned for $x(0) = 0$

1 point is earned for finding value of C

$$\begin{aligned} x(t) &= \int v(t) dt \\ &= \int [1 - \sin(2\pi t)] dt \\ &= t + \frac{1}{2\pi} \cos(2\pi t) + C \end{aligned}$$

$$x(0) = 0 = 0 + \frac{\cos(0)}{2\pi} + C$$

$$\therefore x(t) = t + \frac{1}{2\pi} \cos(2\pi t) - \frac{1}{2\pi}$$

$$1 - \sin(2\pi t) = 0$$

$$t = \frac{1}{4} \text{ or } t = \frac{5}{4}$$



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

2 points are earned for the correct antiderivative of $v(t)$

1 point is earned for $x(0) = 0$

1 point is earned for finding value of C

$$\begin{aligned} x(t) &= \int v(t) dt \\ &= \int [1 - \sin(2\pi t)] dt \\ &= t + \frac{1}{2\pi} \cos(2\pi t) + C \end{aligned}$$

$$x(0) = 0 = 0 + \frac{\cos(0)}{2\pi} + C$$

$$\therefore x(t) = t + \frac{1}{2\pi} \cos(2\pi t) - \frac{1}{2\pi}$$

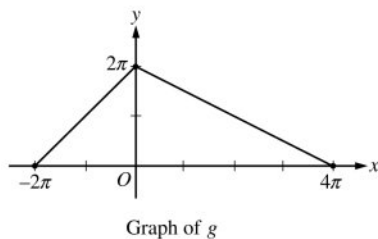
$$1 - \sin(2\pi t) = 0$$

6-8-9

$$t = \frac{1}{4} \text{ or } t = \frac{5}{4}$$

6-8-9

14.



Let g be the piecewise-linear function defined on $[-2\pi, 4\pi]$ whose graph is given above, and let $f(x) = g(x) - \cos\left(\frac{x}{2}\right)$.

- (a) Find $\int_{-2\pi}^{4\pi} f(x)dx$. Show the computations that lead to your answer.
- (b) Find all x -values in the open interval $(-2\pi, 4\pi)$ for which f has a critical point.
- (c) Let $h(x) = \int_0^{3x} g(t)dt$. Find $h'\left(-\frac{\pi}{3}\right)$.

Part A

1 point is earned for the antiderivative

1 point is earned for the answer

$$\begin{aligned}\int_{-2\pi}^{4\pi} f(x)dx &= \int_{-2\pi}^{4\pi} \left(g(x) - \cos\left(\frac{x}{2}\right)\right)dx \\ &= 6\pi^2 - \left[2\sin\left(\frac{x}{2}\right)\right]_{x=-2\pi}^{x=4\pi} \\ &= 6\pi^2\end{aligned}$$

✓

0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the antiderivative

1 point is earned for the answer

$$\begin{aligned}\int_{-2\pi}^{4\pi} f(x)dx &= \int_{-2\pi}^{4\pi} \left(g(x) - \cos\left(\frac{x}{2}\right)\right)dx \\ &= 6\pi^2 - \left[2\sin\left(\frac{x}{2}\right)\right]_{x=-2\pi}^{x=4\pi} \\ &= 6\pi^2\end{aligned}$$

Part B

1 point is earned for $\frac{d}{dx}\left(\cos\left(\frac{x}{2}\right)\right)$

1 point is earned for $g'(x)$

1 point is earned for $x = 0$

1 point is earned for $x = \pi$

6-8-9

$$f'(x) = g'(x) + \frac{1}{2}\sin\left(\frac{x}{2}\right) = \begin{cases} 1 + \frac{1}{2}\sin\left(\frac{x}{2}\right) & \text{for } -2\pi < x < 0 \\ -\frac{1}{2} + \frac{1}{2}\sin\left(\frac{x}{2}\right) & \text{for } 0 < x < 4\pi \end{cases}$$

$f'(x)$ does not exist at $x = 0$.

For $-2\pi < x < 0$, $f'(x) \neq 0$.

For $0 < x < 4\pi$, $f'(x) = 0$ when $x = \pi$.

f has critical points at $x = 0$ and $x = \pi$.



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for $\frac{d}{dx}\left(\cos\left(\frac{x}{2}\right)\right)$

1 point is earned for $g'(x)$

1 point is earned for $x = 0$

1 point is earned for $x = \pi$

$$f'(x) = g'(x) + \frac{1}{2}\sin\left(\frac{x}{2}\right) = \begin{cases} 1 + \frac{1}{2}\sin\left(\frac{x}{2}\right) & \text{for } -2\pi < x < 0 \\ -\frac{1}{2} + \frac{1}{2}\sin\left(\frac{x}{2}\right) & \text{for } 0 < x < 4\pi \end{cases}$$

$f'(x)$ does not exist at $x = 0$.

For $-2\pi < x < 0$, $f'(x) \neq 0$.

For $0 < x < 4\pi$, $f'(x) = 0$ when $x = \pi$.

f has critical points at $x = 0$ and $x = \pi$.

Part C

2 points are earned for $h'(x)$

1 point is earned for the answer

$$h'(x) = g(3x) \cdot 3$$

$$h'\left(-\frac{\pi}{3}\right) = 3g(-\pi) = 3\pi$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for $h'(x)$

1 point is earned for the answer

6-8-9

$$h'(x) = g(3x) \cdot 3$$

$$h'\left(-\frac{\pi}{3}\right) = 3g(-\pi) = 3\pi$$

15. If F is an antiderivative of a continuous function f , then $\int_{-1}^2 f(3x+4)dx =$

(A) $\frac{1}{3}F(10) - \frac{1}{3}F(1)$ ✓

(B) $\frac{1}{3}F(2) - \frac{1}{3}F(-1)$

(C) $F(10) - F(1)$

(D) $F(2) - F(-1)$

Answer A

Correct. Starting with the substitution $u = 3x + 4$, $du = 3dx \Rightarrow dx = \frac{1}{3}du$. When $x = -1$, $u = 1$. When $x = 2$, $u = 10$.

Substituting for $3x + 4$, for dx , and for the limits of integration gives $\int_{-1}^2 f(3x+4)dx = \int_1^{10} f(u)\left(\frac{1}{3}du\right) = \frac{1}{3} \int_1^{10} f(u)du = \frac{1}{3}(F(10) - F(1))$, where the last equality follows from the Fundamental Theorem of Calculus.

Alternatively, since $F(x)$ is an antiderivative of $f(x)$, an antiderivative of $f(3x+4)$ is $\frac{1}{3}F(3x+4)$, where $F(3x+4)$ has been multiplied by $\frac{1}{3}$ to adjust for the chain rule. Therefore $\int_{-1}^2 f(3x+4)dx = \left.\frac{1}{3}F(3x+4)\right|_{-1}^2 = \frac{1}{3}F(10) - \frac{1}{3}F(1)$.

16. $\int_0^3 (x+1)^{1/2}dx =$

(A) $\frac{21}{2}$

(B) 7

(C) $\frac{16}{3}$

(D) $\frac{14}{3}$ ✓

(E) $-\frac{1}{4}$

17. Using the substitution $u = \sqrt{x}$, $\int_1^4 \frac{e^{\sqrt{x}}}{\sqrt{x}}dx$ is equal to which of the following?

(A) $2 \int_1^{16} e^u du$

(B) $2 \int_1^4 e^u du$

(C) $2 \int_1^2 e^u du$ ✓

(D) $\frac{1}{2} \int_1^2 e^u du$

(E) $\int_1^4 e^u du$

6-8-9

18.

If the substitution $\sqrt{x} = \sin y$ is made in the integrand of $\int_0^{1/2} \frac{\sqrt{x}}{\sqrt{1-x}} dx$, the resulting integral is

(A) $\int_0^{1/2} \sin^2 y dy$

(B) $2 \int_0^{1/2} \frac{\sin^2 y}{\cos y} dy$

(C) $2 \int_0^{\pi/4} \sin^2 y dy$ ✓

(D) $\int_0^{\pi/4} \sin^2 y dy$

(E) $2 \int_0^{\pi/6} \sin^2 y dy$

6-8-9

19. The function f is defined by $f(x) = \sqrt{25 - x^2}$ for $-5 \leq x \leq 5$.

(a) Find $f'(x)$.

(b) Write an equation for the line tangent to the graph of f at $x = -3$.

(c) Let g be the function defined by $g(x) = \begin{cases} f(x) & \text{for } -5 \leq x \leq -3 \\ x + 7 & \text{for } -3 < x \leq 5. \end{cases}$

Is g continuous at $x = -3$? Use the definition of continuity to explain your answer.

(d) Find the value of $\int_0^5 x\sqrt{25 - x^2} dx$.

Part A

2 points are earned for $f'(x)$

$$f'(x) = \frac{1}{2}(25 - x^2)^{-1/2}(-2x) = \frac{-x}{\sqrt{25 - x^2}}, \quad -5 < x < 5$$

✓

0	1	2
---	---	---

The student response earns all of the following points:

2 points are earned for $f'(x)$

$$f'(x) = \frac{1}{2}(25 - x^2)^{-1/2}(-2x) = \frac{-x}{\sqrt{25 - x^2}}, \quad -5 < x < 5$$

Part B

1 point is earned for $f'(-3)$

1 point is earned for the answer

$$f'(-3) = \frac{3}{\sqrt{25-9}} = \frac{3}{4}$$

$$f(-3) = \sqrt{25-9} = 4$$

An equation for the tangent line is $y = 4 + \frac{3}{4}(x + 3)$.

✓

0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $f'(-3)$

1 point is earned for the answer

$$f'(-3) = \frac{3}{\sqrt{25-9}} = \frac{3}{4}$$

$$f(-3) = \sqrt{25-9} = 4$$

6-8-9

An equation for the tangent line is $y = 4 + \frac{3}{4}(x + 3)$.

Part C

1 point is earned for considering one-sided limits

1 point is earned for the answer with explanation

$$\lim_{x \rightarrow -3^-} g(x) = \lim_{x \rightarrow -3^-} f(x) = \lim_{x \rightarrow -3^-} \sqrt{25 - x^2} = 4$$

$$\lim_{x \rightarrow -3^+} g(x) = \lim_{x \rightarrow -3^+} (x + 7) = 4$$

Therefore, $\lim_{x \rightarrow -3} g(x) = 4$.

$$g(-3) = f(-3) = 4$$

So, $\lim_{x \rightarrow -3} g(x) = g(-3)$.

Therefore, g is continuous at $x = -3$.

✓

0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for considering one-sided limits

1 point is earned for the answer with explanation

$$\lim_{x \rightarrow -3^-} g(x) = \lim_{x \rightarrow -3^-} f(x) = \lim_{x \rightarrow -3^-} \sqrt{25 - x^2} = 4$$

$$\lim_{x \rightarrow -3^+} g(x) = \lim_{x \rightarrow -3^+} (x + 7) = 4$$

Therefore, $\lim_{x \rightarrow -3} g(x) = 4$.

$$g(-3) = f(-3) = 4$$

So, $\lim_{x \rightarrow -3} g(x) = g(-3)$.

Therefore, g is continuous at $x = -3$.

Part D

2 points are earned for antiderivative

1 point is earned for the answer

$$\text{Let } u = 25 - x^2 \Rightarrow du = -2x dx$$

$$\int_0^5 x \sqrt{25 - x^2} dx = -\frac{1}{2} \int_{25}^0 \sqrt{u} du$$

$$= \left[-\frac{1}{2} \cdot \frac{2}{3} u^{3/2} \right]_{u=25}^{u=0}$$

$$= -\frac{1}{3} (0 - 125) = \frac{125}{3}$$

✓

0	1	2	3
---	---	---	---

6-8-9

The student response earns all of the following points:

2 points are earned for antiderivative

1 point is earned for the answer

$$\text{Let } u = 25 - x^2 \Rightarrow du = -2x dx$$

$$\begin{aligned}\int_0^5 x \sqrt{25 - x^2} dx &= -\frac{1}{2} \int_{25}^0 \sqrt{u} du \\ &= \left[-\frac{1}{2} \cdot \frac{2}{3} u^{3/2} \right]_{u=25}^{u=0} \\ &= -\frac{1}{3} (0 - 125) = \frac{125}{3}\end{aligned}$$

20. If $dy/dx = \tan x$, then $y =$

(A) $12 \tan^2 x + C$

(B) $\sec^2 x + C$

(C) $\ln |\sec x| + C$ ✓

(D) $\ln |\cos x| + C$

(E) $\sec x \tan x + C$

21. Which of the following definite integrals has the same value as $\int_0^4 x e^{x^2} dx$?

(A) $\frac{1}{2} \int_0^4 e^u du$

(B) $\frac{1}{2} \int_0^{16} e^u du$ ✓

(C) $2 \int_0^2 e^u du$

(D) $2 \int_0^4 e^u du$

(E) $2 \int_0^{16} e^u du$

22. The function f is continuous and $\int_0^8 f(u) du = 6$. What is the value of $\int_1^3 x f(x^2 - 1) dx$?

(A) $\frac{3}{2}$

(B) 3 ✓

(C) 6

(D) 12

(E) 24

6-8-9

23. Let f be a function such that $f'(x) = 6x + 8$.

Find $f(x)$ if the graph of f is tangent to the line $3x - y = 2$ at the point $(0, -2)$.

Part A

1 point is earned for $f'(0) = 3$

1 point is earned for finding C

1 point is earned for using f' to find f with constant

1 point is earned for using $(0, -2)$ to find d

$$f'(x) = 3x^2 + 8x + C$$

$$f'(0) = 3$$

$$C = 3$$

$$f(x) = x^3 + 4x^2 + 3x + d$$

$$d = -2$$

$$f(x) = x^3 + 4x^2 + 3x - 2$$



0	1	2	3	4	5
---	---	---	---	---	---

The student response earns five of the following points:

1 point is earned for $f'(x)$ with C

1 point is earned for $f'(0) = 3$

1 point is earned for finding C

1 point is earned for using f' to find f with constant

1 point is earned for using $(0, -2)$ to find d

$$f'(x) = 3x^2 + 8x + C$$

$$f'(0) = 3$$

$$C = 3$$

$$f(x) = x^3 + 4x^2 + 3x + d$$

$$d = -2$$

$$f(x) = x^3 + 4x^2 + 3x - 2$$

24. $\int_1^4 |x - 3| dx =$

(A) $-\frac{3}{2}$

(B) $\frac{3}{2}$

(C) $\frac{5}{2}$

(D) $\frac{9}{2}$

(E) 5



6-8-9

25. $\int_0^1 (3x - 2)^2 dx =$

- (A) $-\frac{7}{3}$
(B) $-\frac{7}{9}$
(C) $\frac{1}{9}$

(D) 1

(E) 3



26. $\int_0^{\frac{\pi}{2}} \frac{\cos \theta}{\sqrt{1 + \sin \theta}} d\theta =$

- (A) $-2(\sqrt{2} - 1)$
(B) $-2\sqrt{2}$
(C) $2\sqrt{2}$

(D) $2(\sqrt{2} - 1)$

(E) $2(\sqrt{2} + 1)$



27. $\int (x^2 + 1)^2 dx =$

- (A) $\frac{(x^2+1)^3}{3} + C$
(B) $\frac{(x^2+1)^3}{6x} + C$
(C) $\left(\frac{x^3}{3} + x\right)^2 + C$
(D) $\frac{2x(x^2+1)^3}{3} + C$

(E) $\frac{x^5}{5} + \frac{2x^3}{3} + x + C$



28. $\int \frac{x^3 + 5}{x^2} dx =$

- (A) $1 - \frac{10}{x^3} + C$
(B) $\frac{3x}{4} + \frac{15}{x^2} + C$

(C) $\frac{x^2}{2} - \frac{5}{x} + C$

(D) $\frac{x^2}{2} - \frac{5}{3x^3} + C$

(E) $-\frac{x^3}{4} - 5 + C$



29. $\int \left(5e^{2x} + \frac{1}{x}\right) dx =$

(A) $\frac{5}{2}e^{2x} + \frac{2}{x^2} + C$

(B) $\frac{5}{2}e^{2x} + \ln|x| + C$

(C) $5e^{2x} + \frac{2}{x^2} + C$

(D) $5e^{2x} + \ln|x| + C$

(E) $10e^{2x} - \frac{1}{x^2} + C$



6-8-9

30. $\int (x^3 - 3x) dx =$

- (A) $3x^2 - 3 + C$
(B) $4x^4 - 6x^2 + C$
(C) $x^4/3 - 3x^2 + C$
(D) $x^4/4 - 3x + C$

(E) $x^4/4 - 3x^2/2 + C$ ✓

31. $\int \frac{1}{x^2} dx =$

- (A) $\ln x^2 + C$
(B) $-\ln x^2 + C$
(C) $x^{-1} + C$

(D) $-x^{-1} + C$ ✓

(E) $-2x^{-3} + C$

32. $\int \frac{x dx}{\sqrt{3x^2 + 5}} =$

- (A) $\frac{1}{9}(3x^2 + 5)^{\frac{3}{2}} + C$
(B) $\frac{1}{4}(3x^2 + 5)^{\frac{3}{2}} + C$
(C) $\frac{1}{12}(3x^2 + 5)^{\frac{1}{2}} + C$

(D) $\frac{1}{3}(3x^2 + 5)^{\frac{1}{2}} + C$ ✓

(E) $\frac{3}{2}(3x^2 + 5)^{\frac{1}{2}} + C$

33. $\int 5x(\sqrt{x} - x^2) dx =$

- (A) $\frac{15\sqrt{x}}{2} - 15x^2 + C$
(B) $5x - \frac{5x^4}{4} + C$

(C) $2x^{5/2} - \frac{5x^4}{4} + C$ ✓

(D) $\frac{25x^{5/2}}{2} - \frac{5x^4}{4} + C$

(E) $\frac{5x^{7/2}}{3} - \frac{5x^6}{6} + C$

34. $\int \sec^2 x dx =$

(A) $\tan x + C$ ✓

(B) $\csc^2 x + C$

(C) $\cos^2 x + C$

(D) $\frac{\sec^3 x}{3} + C$

(E) $2 \sec^2 x \tan x + C$

6-8-9

35. $\int \cos(3x) dx =$

(A) $-3 \sin(3x) + c$

(B) $-\frac{1}{3} \sin(3x) + c$

(C) $\frac{1}{3} \sin(3x) + c$ ✓

(D) $\sin(3x) + c$

(E) $3 \sin(3x) + c$

36. $\int (2^t + e^\pi) dt =$

(A) $\frac{2^{t+1}}{t+1} + \frac{e^{\pi+1}}{\pi+1} + C$

(B) $\frac{2^t}{\ln 2} + e^\pi t + C$ ✓

(C) $\frac{2^t}{\ln 2} + e^\pi + C$

(D) $2^t \ln 2 + \frac{e^{\pi+1}}{\pi+1} + C$

(E) $2^t \ln 2 + e^\pi t + C$

37. Which of the following is an antiderivative of $3\sec^2 x + 2$?

(A) $3 \tan x$

(B) $3 \tan x + 2x$ ✓

(C) $3 \sec x + 2x$

(D) $\sec^3 x + 2x$

(E) $6\sec^2 x + \tan x$

38. $\int_0^2 (x^3 + 1)^{1/2} x^2 dx =$

(A) $\frac{52}{9}$ ✓

(B) 6

(C) $\frac{26}{3}$

(D) $\frac{52}{3}$

39. $\int x^2 (x^3 + 5)^6 dx =$

(A) $\frac{1}{3} (x^3 + 5)^6 + C$

(B) $\frac{1}{3} x^3 \left(\frac{1}{4} x^4 + 5x \right)^6 + C$

(C) $\frac{1}{7} (x^3 + 5)^7 + C$

(D) $\frac{3}{7} x^2 (x^3 + 5)^7 + C$

(E) $\frac{1}{21} (x^3 + 5)^7 + C$ ✓

6-8-9

40. $\int_0^1 x\sqrt{1+8x^2}dx =$

(A) $\frac{1}{24}$

(B) $\frac{13}{12}$ ✓

(C) $\frac{9}{8}$

(D) $\frac{52}{3}$

(E) 18

41. $\int x^2(x^3 - 1)^{10}dx =$

(A) $\frac{x^3}{3}\left(\frac{x^4}{4} - x\right)^{10} + C$

(B) $\frac{(x^3-1)^{11}}{11} + C$

(C) $\frac{x^2(x^3-1)^{11}}{11} + C$

(D) $\frac{(x^3-1)^{11}}{33} + C$ ✓

(E) $\frac{x^3(x^3-1)^{11}}{33} + C$

42. $\int_0^4 \frac{x}{\sqrt{x^2+9}}dx =$

(A) -2

(B) $-\frac{2}{15}$

(C) 1

(D) 2 ✓

(E) 5

43. $\int \frac{1}{3x+12}dx =$

(A) $-3\ln|x+4| + C$

(B) $\frac{1}{3}\ln|x+4| + C$ ✓

(C) $\ln|x+4| + C$

(D) $3\ln|x+4| + C$

44. $\int_0^4 \frac{x}{\sqrt{x^2+9}}dx =$

(A) -2

(B) -2/15

(C) 1

(D) 2 ✓

(E) 5

6-8-9

45. Using the substitution $u = \sin(2x)$, $\int_{\pi/6}^{\pi/2} \sin^5(2x) \cos(2x) dx$ is equivalent to

(A) $-2 \int_{1/2}^1 u^5 du$

(B) $\frac{1}{2} \int_{1/2}^1 u^5 du$

(C) $\frac{1}{2} \int_0^{\sqrt{3}/2} u^5 du$

(D) $\frac{1}{2} \int_{\sqrt{3}/2}^0 u^5 du$ ✓

(E) $2 \int_{\sqrt{3}/2}^0 u^5 du$

46. $\int x e^{x^2} dx =$

(A) $\frac{1}{2} e^{x^2} + C$ ✓

(B) $e^{x^2} + C$

(C) $x e^{x^2} + C$

(D) $\frac{1}{2} e^{2x} + C$

(E) $e^{2x} + C$

47. $\int_0^{\pi/3} \sin(3x) dx =$

(A) -2

(B) -2/3

(C) 0

(D) 2/3 ✓

(E) 2

48. Let f be a function such that $\int_6^{12} f(2x) dx = 10$. Which of the following must be true?

(A) $\int_{12}^{24} f(t) dt = 5$

(B) $\int_{12}^{24} f(t) dt = 20$ ✓

(C) $\int_6^{12} f(t) dt = 5$

(D) $\int_6^{12} f(t) dt = 20$

(E) $\int_3^6 f(t) dt = 5$

49. $\int_2^4 \frac{dx}{5-3x} =$

(A) $-\ln 7$

(B) $-\frac{\ln 7}{3}$ ✓

(C) $\frac{\ln 7}{3}$

(D) $\ln 7$

(E) $3 \ln 7$

6-8-9

50. Using the substitution $u = x + 1$ $\int \frac{x}{\sqrt{x+1}} dx$ is equivalent to

(A) $\int \frac{1}{u+1} du$

(B) $\int u^{-1/2} du$

(C) $\int (u^{1/2} - u^{-1/2}) du$ ✓

(D) $(u-1) \int u^{-1/2} du$

51. $\int 6e^{3x} dx =$

(A) $2e^{3x} + C$ ✓

(B) $6e^{3x} + C$

(C) $18e^{3x} + C$

(D) $\frac{6e^{3x+1}}{3x+1} + C$

52. Using the substitution $u = x^2 - 3$, $\int_{-1}^4 x(x^2 - 3)^5 dx$ is equal to which of the following?

(A) $2 \int_{-2}^{13} u^5 du$

(B) $\int_{-2}^{13} u^5 du$

(C) $\frac{1}{2} \int_{-2}^{13} u^5 du$ ✓

(D) $\int_{-1}^4 u^5 du$

(E) $\frac{1}{2} \int_{-1}^4 u^5 du$

53. $\int x\sqrt{4-x^2} dx =$

(A) $\frac{(4-x^2)^{3/2}}{3} + C$

(B) $-(4-x^2)^{3/2} + C$

(C) $\frac{x^2(4-x^2)^{3/2}}{3} + C$

(D) $-\frac{x^2(4-x^2)^{3/2}}{3} + C$

(E) $-\frac{(4-x^2)^{3/2}}{3} + C$ ✓

54. $\int_0^1 e^{-4x} dx =$

(A) $\frac{-e^{-4}}{4}$

(B) $-4e^{-4}$

(C) $e^{-4} - 1$

(D) $\frac{1}{4} - \frac{e^{-4}}{4}$ ✓

(E) $4 - 4e^{-4}$

6-8-9

55. $\int (3x+1)^5 dx =$

(A) $\frac{(3x+1)^6}{18} + C$ ✓

(B) $\frac{(3x+1)^6}{6} + C$

(C) $\frac{(3x+1)^6}{2} + C$

(D) $\frac{\left(\frac{3x^2}{2} + x\right)^6}{2} + C$

(E) $\left(\frac{3x^2}{2} + x\right)^5 + C$

56. $\int x^2 \cos(x^3) dx =$

(A) $-\frac{1}{3}\sin(x^3) + C$

(B) $\frac{1}{3}\sin(x^3) + C$ ✓

(C) $-\frac{x^3}{3}\sin(x^3) + C$

(D) $\frac{x^3}{3}\sin(x^3) + C$

(E) $\frac{x^3}{3}\sin\left(\frac{x^4}{4}\right) + C$

57. $\int (\sin(2x) + \cos(2x)) dx =$

(A) $\frac{1}{2}\cos(2x) + \frac{1}{2}\sin(2x) + C$

(B) $-\frac{1}{2}\cos(2x) + \frac{1}{2}\sin(2x) + C$ ✓

(C) $2\cos(2x) + 2\sin(2x) + C$

(D) $2\cos(2x) - 2\sin(2x) + C$

(E) $-2\cos(2x) + 2\sin(2x) + C$

58. $\int x^2 \cos(x^3) dx =$

(A) $-\frac{1}{3}\sin(x^3) + C$

(B) $\frac{1}{3}\sin(x^3) + C$ ✓

(C) $-\frac{x^3}{3}\sin(x^3) + C$

(D) $\frac{x^3}{3}\sin(x^3) + C$

(E) $\frac{x^3}{3}\sin\left(\frac{x^4}{4}\right) + C$

6-8-9

59. Using the substitution $u = 2x + 1$, $\int_0^2 \sqrt{2x+1} dx$ is equivalent to

(A) $\frac{1}{2} \int_{-1/2}^{1/2} \sqrt{u} du$

(B) $\frac{1}{2} \int_0^2 \sqrt{u} du$

(C) $\frac{1}{2} \int_1^5 \sqrt{u} du$ ✓

(D) $\int_0^2 \sqrt{u} du$

(E) $\int_1^5 \sqrt{u} du$

60. $\int \frac{x}{x^2 - 4} dx =$

(A) $\frac{-1}{4(x^2 - 4)^2} + C$

(B) $\frac{1}{2(x^2 - 4)} + C$

(C) $\frac{1}{2} \ln |x^2 - 4| + C$ ✓

(D) $2 \ln |x^2 - 4| + C$

(E) $\frac{1}{2} \arctan\left(\frac{x}{2}\right) + C$

61. $\int \frac{e^{\sqrt{x}}}{\sqrt{x}} dx$

(A) $2e^{\sqrt{x}} + c$ ✓

(B) $\frac{1}{2}e^{\sqrt{x}} + c$

(C) $e^{\sqrt{x}} + c$

(D) $2\sqrt{x}e^{\sqrt{x}} + c$

(E) $\frac{1}{2} \frac{e^{\sqrt{x}}}{\sqrt{x}} + c$

62. $\int (x^3 + 1)^2 dx =$

(A) $\frac{1}{7}x^7 + x + c$

(B) $\frac{1}{7}x^7 + \frac{1}{2}x^4 + x + c$ ✓

(C) $6x^2(x^3 + 1) + c$

(D) $\frac{1}{3}(x^3 + 1)^3 + c$

(E) $\frac{(x^3 + 1)^3}{9x^2} + c$

6-8-9

63.
$$\int_0^{\ln 2} \frac{e^x}{1+(e^x-1)^2} dx =$$

(A) $\arctan(\ln 2)$

(B) $\ln 2$

(C) $\pi/4$

(D) $\pi/2$



Answer C

This option is correct. This integral can be evaluated using substitution of variables. Let $u = e^x - 1$ since that is the term being squared in the denominator.

$$u = e^x - 1 \Rightarrow du = e^x dx$$

When $x = 0$, $u = e^0 - 1 = 0$. When $x = \ln 2$, $u = e^{\ln 2} - 1 = 2 - 1 = 1$.

The definite integral becomes $\int_0^1 \frac{1}{1+u^2} du$. The integrand should then be recognized as the derivative of the arctan function:

$$\int_0^1 \frac{1}{1+u^2} du = \tan^{-1} u \Big|_0^1 = \tan^{-1}(1) - \tan^{-1}(0) = \frac{\pi}{4} - 0 = \frac{\pi}{4}.$$

64.
$$\int \frac{e^x}{1+e^x} dx$$

(A) $\ln\left(\frac{1}{e^x} + 1\right) + c$

(B) $\ln(1+e^x) + c$

(C) $x - \ln(1+e^x) + c$

(D) $e^x + x + c$

(E) $\tan^{-1}(e^x) + c$



65. If $\int_1^2 f(x-c) dx = 5$ where c is a constant, then $\int_{1-c}^{2-c} f(x) dx =$

(A) $5+c$

(B) 5

(C) $5-c$

(D) $c-5$

(E) -5



6-8-9

66. Let g be a continuous function. Using the substitution $u = 2x - 1$, the integral $\int_2^3 g(2x - 1) dx$ is equal to which of the following?

(A) $\int_2^3 g(u) du$

(B) $\frac{1}{2} \int_2^3 g(u) du$

(C) $\int_3^5 g(u) du$

(D) $\frac{1}{2} \int_3^5 g(u) du$ ✓

Answer D

This option is correct. Starting with the substitution $u = 2x - 1$,

$$u = 2x - 1 \Rightarrow du = 2dx \Rightarrow dx = \frac{1}{2} du.$$

Also change the limits of integration for x to limits of integration for u :

$$x = 2 \Rightarrow u = 2 \cdot 2 - 1 = 3$$

$$x = 3 \Rightarrow u = 2 \cdot 3 - 1 = 5.$$

Substituting for $2x - 1$, for dx , and for the limits of integration gives:

$$\int_2^3 g(2x - 1) dx = \int_3^5 g(u) \cdot \frac{1}{2} du = \frac{1}{2} \int_3^5 g(u) du$$

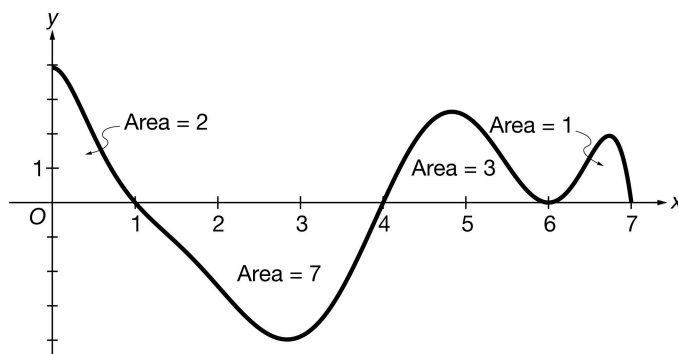
6-8-9

67. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of f'

The figure above shows the graph of f' , the derivative of a differentiable function f , on the closed interval $0 \leq x \leq 7$. The areas of the regions between the graph of f' and the x -axis are labeled in the figure. The function f is defined for all real numbers and satisfies $f(4) = 10$.

Let g be the function defined by $g(x) = 5 - x^2$.

(a) Find the value of $\int_0^7 f'(x) \, dx$.

(b) Given that $f(4) = 10$, write an expression for $f(x)$ that involves an integral. Use this expression to find the absolute minimum value of f and the absolute maximum value of f on the closed interval $0 \leq x \leq 7$. Justify your answers.

(c) Find $\int g(x) \, dx$.

(d) Find the value of $\int_1^2 x f'(g(x)) \, dx$.

Part A

The response should include supporting work for the numerical answer of -1 .

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1
---	---

The student response accurately includes a correct value.

Solution:

$$\int_0^7 f'(x) \, dx = 2 - 7 + 3 + 1 = -1$$

Part B

The first point is earned for a correct expression for $f(x)$ OR evidence of correct use of FTC in determining the appropriate $f(x)$ values for consideration.

For the second point: The response may also consider $x = 6$, though this is not required in order to justify an absolute minimum value or an absolute maximum value on the interval.

6-8-9

For the third and fourth points: At most 1 out of 2 points is earned for supporting work that justifies a correct absolute minimum value OR absolute maximum value based on a Candidates Test with a maximum of one arithmetic error.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3	4
---	---	---	---	---

The student response accurately includes all four of the criteria below.

- ☐ use of Fundamental Theorem of Calculus
- ☐ considers $x = 1$ and $x = 4$
- ☐ answers for absolute minimum value and absolute maximum value
- ☐ justification

Solution:

$$f(x) = f(4) + \int_4^x f'(t) dt$$

The absolute minimum and absolute maximum will occur at a critical point where $f'(x) = 0$ or at an endpoint.

$$f'(x) = 0 \Rightarrow x = 1, x = 4, x = 6$$

The critical point at $x = 6$ is neither the location of an absolute minimum nor an absolute maximum because f' does not change sign at $x = 6$. Thus, the only candidates are $x = 0$, $x = 1$, $x = 4$, and $x = 7$.

x	$f(x)$
0	$f(4) + \int_4^0 f'(t) dt = f(4) - \int_0^4 f'(t) dt$ $= 10 - (2 - 7) = 15$
1	$f(4) + \int_4^1 f'(t) dt = f(4) - \int_1^4 f'(t) dt$ $= 10 - (-7) = 17$
4	10
7	$f(4) + \int_4^7 f'(t) dt = 10 + (3 + 1) = 14$

On the interval $0 \leq x \leq 7$, the absolute minimum value is $f(4) = 10$ and the absolute maximum value is $f(1) = 17$.

Part C

A response with any errors, including omission of the constant of integration, does not earn the point.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1
---	---

The student response accurately includes a correct answer.

Solution:

$$\int g(x) dx = \int (5 - x^2) dx = 5x - \frac{1}{3}x^3 + C$$

Part D

6-8-9

To earn the second point, a response must present a definite integral that correctly handles the substitution of the limits of integration. Correct handling of the reversal of the limits of integration and all calculations are part of the third point.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2	3
---	---	---	---

The student accurately includes all three of the criteria below.

- ☐ sets $u = 5 - x^2$
- ☐ integral in terms of u
- ☐ answer

Solution:

Let $u = g(x) = 5 - x^2$.

Then $du = -2x \, dx$ and $-\frac{1}{2} du = x \, dx$.

$$x = 2 \Rightarrow u = 5 - 2^2 = 1$$

$$x = 1 \Rightarrow u = 5 - 1^2 = 4$$

$$\int_1^2 x f'(g(x)) \, dx = -\frac{1}{2} \int_4^1 f'(u) \, du = \frac{1}{2} \int_1^4 f'(u) \, du = \frac{1}{2}(-7) = -\frac{7}{2}$$

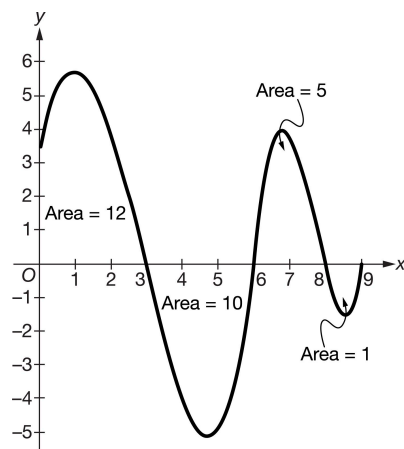
6-8-9

68. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of f'

The figure above shows the graph of f' , the derivative of a differentiable function f , on the closed interval $0 \leq x \leq 9$. The areas of the regions between the graph of f' and the x -axis are labeled in the figure. The function f is defined for all real numbers and satisfies $f(6) = 7$.

Let g be the function defined by $g(x) = x^2 - 1$.

- Find the value of $\int_0^9 f'(x) \, dx$.
- Given that $f(6) = 7$, write an expression for $f(x)$ that involves an integral. Use this expression to find the absolute minimum value of f and the absolute maximum value of f on the closed interval $0 \leq x \leq 9$. Justify your answers.
- Find $\int g(x) \, dx$.
- Find the value of $\int_2^3 x f'(g(x)) \, dx$.

Part A

The response should include supporting work for the numerical answer of 6.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The student response accurately includes a correct value.

Solution:

$$\int_0^9 f'(x) \, dx = 12 - 10 + 5 - 1 = 6$$

Part B

The first point is earned for a correct expression for $f(x)$ OR evidence of correct use of FTC undefined in determining the appropriate $f(x)$ values for consideration.

6-8-9

For the third and fourth points: At most 1 out of 2 points is earned for supporting work that justifies a correct absolute minimum value OR absolute maximum value based on a Candidates Test with a maximum of one arithmetic error.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response



0	1	2	3	4
---	---	---	---	---

The student response accurately includes all four of the criteria below.

- ☐ use of Fundamental Theorem of Calculus
- ☐ considers $x = 3$, $x = 6$, and $x = 8$
- ☐ answers for absolute minimum value and absolute maximum value
- ☐ justification

Solution:

$$f(x) = f(6) + \int_6^x f'(t) dt$$

The absolute minimum and absolute maximum will occur at a critical point where $f'(x) = 0$

$$f'(x) = 0 \Rightarrow x = 3, x = 6, x = 8$$

The candidates are $x = 0$, $x = 3$, $x = 6$, $x = 8$, and $x = 9$.

x	$f(x)$
0	$f(6) + \int_6^0 f'(t) dt = f(6) - \int_0^6 f'(t) dt$ $= 7 - (12 - 10) = 5$
3	$f(6) + \int_6^3 f'(t) dt = f(6) - \int_3^6 f'(t) dt$ $= 7 - (-10) = 17$
6	7
8	$f(6) + \int_6^8 f'(t) dt = 7 + 5 = 12$
9	$f(6) + \int_6^9 f'(t) dt = 7 + (5 - 1) = 11$

On the interval $0 \leq x \leq 9$, the absolute minimum value is $f(0) = 5$ and the absolute maximum value is $f(3) = 17$.

Part C

A response with any errors, including omission of the constant of integration, does not earn the point.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1
---	---

The student response accurately includes a correct answer.

Solution:

$$\int g(x) dx = \int (x^2 - 1) dx = \frac{1}{3}x^3 - x + C$$

Part D

6-8-9

To earn the second point, a response must present a definite integral that correctly handles the substitution of the limits of integration. Correct handling of the reversal of the limits of integration and all calculations are part of the third point.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2	3
---	---	---	---

The student accurately includes all three of the criteria below.

- ☐ sets $u = x^2 - 1$
- ☐ integral in terms of u
- ☐ answer

Solution:

Let $u = g(x) = x^2 - 1$.

Then $du = 2x \, dx$ and $\frac{1}{2} du = x \, dx$.

$$x = 3 \Rightarrow u = 3^2 - 1 = 8$$

$$x = 2 \Rightarrow u = 2^2 - 1 = 3$$

$$\int_2^3 x f'(g(x)) \, dx = \frac{1}{2} \int_3^8 f'(u) \, du = \frac{1}{2}(-10 + 5) = -\frac{5}{2}$$

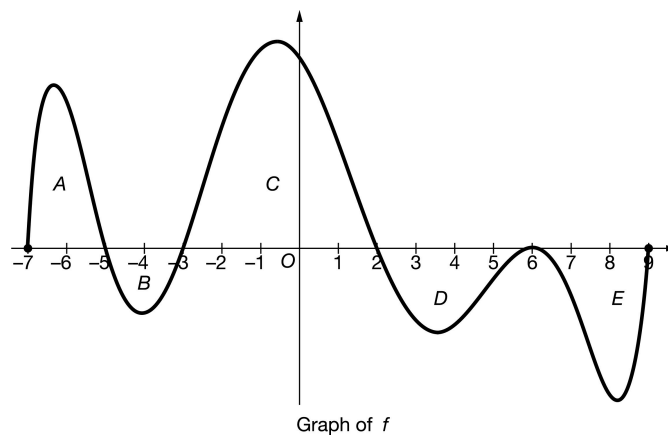
6-8-9

69. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



The figure above shows the graph of the continuous function f . The regions A , B , C , D , and E have areas 5, 2, 16, 5, and 6, respectively. For $-7 \leq x \leq 9$, the function g is defined by $g(x) = -6 + \int_{-3}^x f(t) \, dt$.

- Is there a value of x , for $-3 \leq x \leq 2$, such that $g(x) = 0$? Justify your answer.
- Find the absolute minimum value of g and the absolute maximum value of g on the interval $-7 \leq x \leq 9$. Justify your answer.
- Find $\int (2x + 16) \, dx$.
 - Find the value of $\int_{-7}^{-5} f(2x + 16) \, dx$.

Part A

For the second point, the response must include that g is continuous on $-3 \leq x \leq 2$, show that $g(-3) < 0 < g(2)$, and draw the conclusion from the Intermediate Value Theorem.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ $g(-3)$ and $g(2)$
- ☐ justification, using Intermediate Value Theorem

Solution:

$g'(x) = f(x)$ and f is continuous on $-7 \leq x \leq 9$. Because g is an antiderivative of a continuous function f , g is continuous on $-3 \leq x \leq 2$.

$$g(-3) = -6 + \int_{-3}^{-3} f(t) \, dt = -6$$

$$g(2) = -6 + \int_{-3}^2 f(t) \, dt = -6 + 16 = 10$$

6-8-9

Since $g(-3) < 0 < g(2)$, by the Intermediate Value Theorem there is an x in the interval $[-3, 2]$ such that $g(x) = 0$.

Part B

For the second point: The response may also consider $x = 6$, though this is not required in order to justify an absolute minimum value or an absolute maximum value on the interval.

For the third and fourth points: At most 1 out of 2 points is earned for supporting work that justifies a correct absolute minimum value OR absolute maximum value based on a Candidates Test with a maximum of one arithmetic error.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3	4
---	---	---	---	---

The student response accurately includes all four of the criteria below.

- ☐ $g'(x) = f(x)$
- ☐ considers $x = -5$, $x = -3$, $x = 2$ as candidates
- ☐ answers for absolute minimum value and absolute maximum value
- ☐ justification

Solution:

$g'(x) = f(x)$ changes sign only at $x = -5$, $x = -3$, and $x = 2$.

x	$g(x)$
-7	$-6 + \int_{-3}^{-7} f(t) dt = -6 - \int_{-7}^{-3} f(t) dt = -6 - (5 - 2) = -9$
-5	$-6 + \int_{-3}^{-5} f(t) dt = -6 - \int_{-5}^{-3} f(t) dt = -6 - (-2) = -4$
-3	-6
2	10
9	$-6 + \int_{-3}^9 f(t) dt = -6 + (16 - 5 - 6) = -1$

On the interval $-7 \leq x \leq 9$, the absolute minimum value is $g(-7) = -9$ and the absolute maximum value is $g(2) = 10$.

Part C

A response with any errors, including omission of the constant of integration, does not earn the first point.

For the second point, the response must present a definite integral that correctly handles the substitution of the limits of integration.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ expression for $\int (2x + 16) dx$
- ☐ integral in terms of u
- ☐ answer

Solution:

6-8-9

(i)

$$\int (2x + 16) \, dx = x^2 + 16x + C$$

(ii)

Let $u = 2x + 16$. Then $du = 2 \, dx$ and $\frac{1}{2} \, du = dx$.

$$x = -7 \Rightarrow u = 2(-7) + 16 = 2$$

$$x = -5 \Rightarrow u = 2(-5) + 16 = 6$$

$$\int_{-7}^{-5} f(2x + 16) \, dx = \frac{1}{2} \int_2^6 f(u) \, du = \frac{1}{2}(-5) = -\frac{5}{2}$$

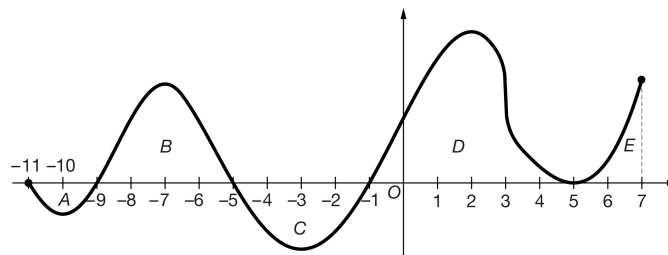
6-8-9

70. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



The figure above shows the graph of the continuous function f . The regions A , B , C , D , and E have areas 2, 14, 10, 27, and 4, respectively. For $-11 \leq x \leq 7$, the function g is defined by $g(x) = 8 + \int_{-5}^x f(t) \, dt$.

- Is there a value of x , for $-5 \leq x \leq -1$, such that $g(x) = 0$? Justify your answer.
- Find the absolute minimum value of g and the absolute maximum value of g on the interval $-11 \leq x \leq 7$. Justify your answer.
- Find $\int (2x + 1) \, dx$.
 - Find the value of $\int_{-5}^{-1} f(2x + 1) \, dx$.

Part A

For the second point, the response must include that g is continuous on $-5 \leq x \leq -1$ show that $g(-1) < 0 < g(-5)$ and draw the conclusion from the Intermediate Value Theorem.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ $g(-5)$ and $g(-1)$
- ☐ justification, using Intermediate Value Theorem

Solution:

$g'(x) = f(x)$ and f is continuous on $-11 \leq x \leq 7$. Because g is an antiderivative of a continuous function f , g is continuous on $-5 \leq x \leq -1$.

$$g(-5) = 8 + \int_{-5}^{-5} f(t) \, dt = 8$$

$$g(-1) = 8 + \int_{-5}^{-1} f(t) \, dt = 8 - 10 = -2$$

Since $g(-1) < 0 < g(-5)$, by the Intermediate Value Theorem there is an x in the interval $[-5, -1]$ such that $g(x) = 0$.

Part B

6-8-9

For the third and fourth points: At most 1 out of 2 points is earned for supporting work that justifies a correct absolute minimum value OR absolute maximum value based on a Candidates Test with a maximum of one arithmetic error.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3	4
---	---	---	---	---

The student response accurately includes all four of the criteria below.

- ☐ $g'(x) = f(x)$
- ☐ considers $x = -9$, $x = -5$, $x = -1$ as candidates
- ☐ answers for absolute minimum value and absolute maximum value
- ☐ justification

Solution:

$g'(x) = f(x)$ changes sign only at $x = -9$, $x = -5$, and

x	$g(x)$
-11	$8 + \int_{-5}^{-11} f(t) dt = 8 - \int_{-11}^{-5} f(t) dt = 8 - (-2 + 14) = -4$
-9	$8 + \int_{-5}^{-9} f(t) dt = 8 - \int_{-9}^{-5} f(t) dt = 8 - 14 = -6$
-5	8
-1	-2
7	$8 + \int_{-5}^7 f(t) dt = 8 + (-10 + 27 + 4) = 29$

On the interval $-11 \leq x \leq 7$, the absolute minimum value is $g(-9) = -6$ and the absolute maximum value is $g(7) = 29$.

Part C

A response with any errors, including omission of the constant of integration, does not earn the first point.

For the second point, the response must present a definite integral that correctly handles the substitution of the limits of integration.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ expression for $\int (2x + 1) dx$
- ☐ integral in terms of u
- ☐ answer

Solution:

(i)

$$\int (2x + 1) dx = x^2 + x + C$$

(ii)

6-8-9

Let $u = 2x + 1$. Then $du = 2 \, dx$ and $\frac{1}{2} \, du = dx$.

$$x = -1 \Rightarrow u = 2(-1) + 1 = -1$$

$$x = -5 \Rightarrow u = 2(-5) + 1 = -9$$

$$\int_{-5}^{-1} f(2x + 1) \, dx = \frac{1}{2} \int_{-9}^{-1} f(u) \, du = \frac{1}{2}(14 - 10) = 2$$

71. $\int e^{-2t} \, dt =$

(A) $e^{-2t} + C$

(B) $-e^{-t} + C$

(C) $-2e^{-2t} + C$

(D) $-\frac{1}{2}e^{-2t} + C$

**Answer D**

Correct. Starting with the substitution $u = -2t$, $u = -2t \Rightarrow \frac{du}{dt} = -2 \Rightarrow dt = -\frac{1}{2}du$. Substituting for $-2t$ and dt gives

$$\int e^{-2t} \, dt = \int e^u \left(-\frac{1}{2}du\right) = -\frac{1}{2} \int e^u \, du = -\frac{1}{2} \cdot e^u + C = -\frac{1}{2}e^{-2t} + C.$$

Alternatively, an antiderivative of e^t is e^t . Therefore an antiderivative of e^{-2t} is $-\frac{1}{2}e^{-2t}$ where the exponential term has been multiplied by $-\frac{1}{2}$ to adjust for the chain rule. Therefore $\int e^{-2t} \, dt = -\frac{1}{2}e^{-2t} + C$.

6-8-9

72. Let f be a function such that $f'(x) = 6x + 8$.

- (a) Find $f(x)$ if the graph of f is tangent to the line $3x - y = 2$ at the point $(0, -2)$.
 (b) Find the average value of $f(x)$ on the closed interval $[-1, 1]$.

Part A

1 point is earned for $f'(x)$ with C

1 point is earned for $f'(0) = 3$

1 point is earned for finding C

1 point is earned for using f' to find f with constant

1 point is earned for using $(0, -2)$ to find d

$$f'(x) = 3x^2 + 8x + C$$

$$f'(0) = 3$$

$$C = 3$$

$$f(x) = x^3 + 4x^2 + 3x + d$$

$$d = -2$$

$$f(x) = x^3 + 4x^2 + 3x - 2$$



0	1	2	3	4	5
---	---	---	---	---	---

The student response earns all of the following points:

1 point is earned for $f'(x)$ with C

1 point is earned for $f'(0) = 3$

1 point is earned for finding C

1 point is earned for using f' to find f with constant

1 point is earned for using $(0, -2)$ to find d

$$f'(x) = 3x^2 + 8x + C$$

$$f'(0) = 3$$

$$C = 3$$

$$f(x) = x^3 + 4x^2 + 3x + d$$

$$d = -2$$

$$f(x) = x^3 + 4x^2 + 3x - 2$$

Part B

3 points are earned for integrand, constant and limits.

2 points are **deducted** for incorrect integrand or incorrect constant

1 point is **deducted** for incorrect limits

1 point is earned for the antidifferentiation and evaluation

6-8-9

$$\begin{aligned}
 & \frac{1}{1-(-1)} \int_{-1}^1 (x^3 + 4x^2 + 3x - 2) dx \\
 &= \frac{1}{2} \left[\frac{1}{4}x^4 + \frac{4}{3}x^3 + \frac{3}{2}x^2 - 2x \right]_{-1}^1 \\
 &= \frac{1}{2} \left[\left(\frac{1}{4} + \frac{4}{3} + \frac{3}{2} - 2 \right) - \left(\frac{1}{4} - \frac{4}{3} + \frac{3}{2} + 2 \right) \right] \\
 &= \boxed{-\frac{2}{3}}
 \end{aligned}$$

✓

0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

3 points are earned for integrand, constant and limits.

2 points are **deducted** for incorrect integrand or incorrect constant

1 point is **deducted** for incorrect limits

1 point is earned for the antidifferentiation and evaluation

$$\begin{aligned}
 & \frac{1}{1-(-1)} \int_{-1}^1 (x^3 + 4x^2 + 3x - 2) dx \\
 &= \frac{1}{2} \left[\frac{1}{4}x^4 + \frac{4}{3}x^3 + \frac{3}{2}x^2 - 2x \right]_{-1}^1 \\
 &= \frac{1}{2} \left[\left(\frac{1}{4} + \frac{4}{3} + \frac{3}{2} - 2 \right) - \left(\frac{1}{4} - \frac{4}{3} + \frac{3}{2} + 2 \right) \right] \\
 &= \boxed{-\frac{2}{3}}
 \end{aligned}$$

6-8-9

73. A particle moves along the x -axis in such a way that its acceleration at time t for $t \geq 0$ is given by $a(t) = 4\cos(2t)$. At time $t = 0$, the velocity of the particle is $v(0) = 1$ and its position is $x(0) = 0$.

- (a) Write an equation for the velocity $v(t)$ of the particle.
 (b) Write an equation for the position $x(t)$ of the particle.
 (c) For what values of t , $0 \leq t \leq \pi$, is the particle at rest?

Part A

1 point is earned for integrand

1 point is earned for $v(t) = 2\sin 2t + C$ or $v(t) = 2\sin 2t$

1 point is earned for answer

$$\begin{aligned} v(t) &= \int 4 \cos 2t \, dt \\ \therefore v(t) &= 2 \sin 2t + C \\ v(0) &= 1 \Rightarrow C = 1 \\ \therefore v(t) &= 2 \sin 2t + 1 \end{aligned}$$

✓

0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for integrand

1 point is earned for $v(t) = 2\sin 2t + C$ or $v(t) = 2\sin 2t$

1 point is earned for answer

$$\begin{aligned} v(t) &= \int 4 \cos 2t \, dt \\ \therefore v(t) &= 2 \sin 2t + C \\ v(0) &= 1 \Rightarrow C = 1 \\ \therefore v(t) &= 2 \sin 2t + 1 \end{aligned}$$

Part B

1 point is earned for integrand

1 point is earned for $x(t) = -\cos 2t + t + C$ or $x(t) = -\cos 2t + t$

1 point is earned for answer

$$\begin{aligned} x(t) &= \int 2 \sin 2t + 1 \, dt \\ \therefore x(t) &= -\cos 2t + t + C \\ x(0) &= 0 \Rightarrow C = 1 \\ \therefore x(t) &= -\cos 2t + t + 1 \end{aligned}$$

✓

0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for integrand

6-8-9

1 point is earned for $x(t) = -\cos 2t + t + C$ or $x(t) = -\cos 2t + t$

1 point is earned for answer

$$x(t) = \int 2\sin 2t + 1 \, dt$$

$$\therefore x(t) = -\cos 2t + t + C$$

$$x(0) = 0 \Rightarrow C = 1$$

$$\therefore x(t) = -\cos 2t + t + 1$$

Part C

1 point is earned if sets $v(t) = 0$

2 point are earned for solving for t

$$2 \sin 2t + 1 = 0$$

$$\sin 2t = -\frac{1}{2}$$

$$\therefore 2t = \frac{7\pi}{6}, \frac{11\pi}{6}$$

$$\therefore t = \frac{7\pi}{12}, \frac{11\pi}{12}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned if sets $v(t) = 0$

2 point are earned for solving for t

$$2 \sin 2t + 1 = 0$$

$$\sin 2t = -\frac{1}{2}$$

$$\therefore 2t = \frac{7\pi}{6}, \frac{11\pi}{6}$$

$$\therefore t = \frac{7\pi}{12}, \frac{11\pi}{12}$$

6-8-9

74. A particle moves along the x -axis so that its velocity at any time $t \geq 0$ is given by $v(t) = 1 - \sin(2\pi t)$.

- (a) Find the acceleration $a(t)$ of the particle at any time t .
 (b) Find all values of t , $0 \leq t \leq 2$, for which the particle is at rest.
 (c) Find the position $x(t)$ of the particle at any time t if $x(0) = 0$.

Part A

2 points are earned for correct differentiation of velocity

$$\begin{aligned} a(t) &= v'(t) \\ &= -2\pi \cos(2\pi t) \end{aligned}$$

✓

0	1	2
---	---	---

The student response earns all of the following points:

2 points are earned for correct differentiation of velocity

$$\begin{aligned} a(t) &= v'(t) \\ &= -2\pi \cos(2\pi t) \end{aligned}$$

Part B

1 point is earned for $1 - \sin(2\pi t) = 0$

1 point is earned for $t = \frac{1}{4}$

1 point is earned for $t = \frac{5}{4}$

$$v(t) = 0$$

$$1 - \sin(2\pi t) = 0 \text{ or } 1 = \sin(2\pi t)$$

$$2\pi t = \frac{\pi}{2} + 2k\pi,$$

where $k = 0, \pm 1, \pm 2, \dots$

$$\text{and } 0 \leq t \leq 2$$

$$\therefore t = \frac{1}{4} \text{ and } t = \frac{5}{4}$$

✓

0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for $1 - \sin(2\pi t) = 0$

6-8-9

1 point is earned for $t = \frac{1}{4}$

1 point is earned for $t = \frac{5}{4}$

$$v(t) = 0$$

$$1 - \sin(2\pi t) = 0 \text{ or } 1 = \sin(2\pi t)$$

$$2\pi t = \frac{\pi}{2} + 2k\pi,$$

where $k = 0, \pm 1, \pm 2, \dots$

$$\text{and } 0 \leq t \leq 2$$

$$\therefore t = \frac{1}{4} \text{ and } t = \frac{5}{4}$$

Part C

2 points are earned for the correct antiderivative of $v(t)$

1 point is earned for $x(0) = 0$

1 point is earned for finding value of C

$$\begin{aligned} x(t) &= \int v(t) dt \\ &= \int [1 - \sin(2\pi t)] dt \\ &= t + \frac{1}{2\pi} \cos(2\pi t) + C \end{aligned}$$

$$x(0) = 0 = 0 + \frac{\cos(0)}{2\pi} + C$$

$$\therefore x(t) = t + \frac{1}{2\pi} \cos(2\pi t) - \frac{1}{2\pi}$$



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

2 points are earned for the correct antiderivative of $v(t)$

1 point is earned for $x(0) = 0$

1 point is earned for finding value of C

6-8-9

$$\begin{aligned}x(t) &= \int v(t) dt \\&= \int [1 - \sin(2\pi t)] dt \\&= t + \frac{1}{2\pi} \cos(2\pi t) + C\end{aligned}$$

$$x(0) = 0 = 0 + \frac{\cos(0)}{2\pi} + C$$

$$\therefore x(t) = t + \frac{1}{2\pi} \cos(2\pi t) - \frac{1}{2\pi}$$

75. $\int \left(\frac{1}{2x} + 2^x \right) dx =$
- (A) $-\frac{1}{2x^2} + 2^x \ln 2 + C$
- (B) $\frac{1}{2} \ln |x| + \frac{2^{x+1}}{x+1} + C$
- (C) $\frac{1}{2} \ln |x| + \frac{2^x}{\ln 2} + C$ ✓
- (D) $\ln |2x| + \frac{2^x}{\ln 2} + C$

Answer C

Correct. $\int \left(\frac{1}{2x} + 2^x \right) dx = \frac{1}{2} \int \frac{1}{x} dx + \int 2^x dx = \frac{1}{2} \ln |x| + \frac{2^x}{\ln 2} + C$

76. $\int \frac{x^2}{4} dx =$
- (A) $\frac{x}{2} + C$
- (B) $\frac{x^3}{12} + C$ ✓
- (C) $\frac{x^3}{4} + C$
- (D) $\frac{3x^3}{4} + C$

Answer B

Correct. By the power rule for antiderivatives, the antiderivative of x^n is $\frac{x^{n+1}}{n+1}$ for $n \neq -1$. Therefore, $\int \frac{x^2}{4} dx = \frac{1}{4} \int x^2 dx = \frac{1}{4} \cdot \frac{x^3}{3} + C = \frac{x^3}{12} + C$.

6-8-9

77. $\int (2x + 1)^8 dx =$

(A) $\frac{(2x+1)^9}{18} + C$

(B) $\frac{(2x+1)^9}{9} + C$

(C) $\frac{2(2x+1)^9}{9} + C$

(D) $\frac{(x^2+x)^9}{9} + C$

**Answer A**

Correct. Starting with the substitution $u = 2x + 1$, $u = 2x + 1 \Rightarrow \frac{du}{dx} = 2 \Rightarrow dx = \frac{1}{2} du$.

Substituting for $2x + 1$ and for dx gives $\int (2x + 1)^8 dx = \int u^8 \left(\frac{1}{2} du \right) = \frac{1}{2} \int u^8 du = \frac{1}{2} \cdot \frac{1}{9} u^9 + C = \frac{1}{18} (2x + 1)^9 + C$.

78. $\int \frac{1}{x^2 + 4} dx =$

(A) $\ln(x^2 + 4) + C$

(B) $\arctan\left(\frac{x}{4}\right) + C$

(C) $\frac{1}{4} \arctan\left(\frac{x}{2}\right) + C$

(D) $\frac{1}{2} \arctan\left(\frac{x}{2}\right) + C$

**Answer D**

Correct. This integral can be found using substitution of variables. The integrand should be recognized as coming from the derivative of an inverse tangent function. Let $u = \frac{x}{2}$, since that is the term being squared in the denominator after removing a factor of 4. Then $du = \frac{1}{2} dx$. The integral becomes the following.

$$\int \frac{1}{x^2 + 4} dx = \frac{1}{4} \int \frac{1}{\frac{x^2}{4} + 1} dx = \frac{1}{4} \int \frac{1}{u^2 + 1} (2du) = \frac{1}{2} \arctan u + C = \frac{1}{2} \arctan\left(\frac{x}{2}\right) + C$$

6-8-9

79.



x	0	1	2	3	4
$g(x)$	1	$\frac{1}{8}$	$-\frac{3}{4}$	$-\frac{13}{8}$	$-\frac{5}{2}$

Selected values of the twice-differentiable function g are given in the table above. What is the value of $\int_0^4 g'(x) \arctan^2(2g(x) + 3) dx$?

(A) -14.588 (B) -3.647 (C) -0.721 (D) 2.136 **Answer B**

Correct. Use the technique of substitution of variables with $u = 2g(x) + 3$.

$$u = 2g(x) + 3 \Rightarrow du = 2g'(x)dx \Rightarrow g'(x)dx = \frac{du}{2}$$

When $x = 0$, $u = 2g(0) + 3 = 2(1) + 3 = 5$.

When $x = 4$, $u = 2g(4) + 3 = 2(-\frac{5}{2}) + 3 = -2$.

$$\int_0^4 g'(x) \arctan^2(2g(x) + 3) dx = \frac{1}{2} \int_5^{-2} \arctan^2 u du = -3.647$$

The calculator is used to do the numerical integration of the definite integral.

80. The function f is continuous, and g is a function such that $g'(x) = f(x)$. Which of the following is equal to $\int_6^{10} f(\frac{1}{2}x) dx$ in terms of g ?

(A) $\frac{1}{2}g(5) - \frac{1}{2}g(3)$ (B) $\frac{1}{2}g(10) - \frac{1}{2}g(6)$ (C) $2g(5) - 2g(3)$ (D) $2g(10) - 2g(6)$ **Answer C**

Correct. This integral can be evaluated using the Fundamental Theorem of Calculus and substitution of variables with $u = \frac{1}{2}x$.

$$u = \frac{1}{2}x \Rightarrow du = \frac{1}{2}dx \Rightarrow dx = 2du$$

When $x = 6$, $u = 3$.

When $x = 10$, $u = 5$.

Substituting for $\frac{1}{2}x$, for dx , and for the limits of integration gives

$$\int_6^{10} f\left(\frac{1}{2}x\right) dx = 2 \int_3^5 f(u) du = 2 \int_3^5 g'(u) du = 2(g(5) - g(3)),$$

where the last equality comes from the Fundamental Theorem of Calculus.

6-8-9

81. Using the substitution $u = x - 1$, $\int_2^{10} 3x\sqrt{x-1} \, dx$ is equivalent to

(A) $\int_1^9 3(u+1)\sqrt{u} \, du$

(B) $\int_1^9 3u(u+1) \, du$

(C) $\int_2^{10} (3u+1)\sqrt{u} \, du$

(D) $\int_2^{10} 3u\sqrt{u+1} \, du$

Answer A

Correct. With the substitution $u = x - 1$, $du = dx$.

When $x = 2$, $u = 1$.

When $x = 10$, $u = 9$.

The substitution $u = x - 1$ is made inside the square root, and the substitution $x = u + 1$ is made for the x term outside the square root. Substituting for the limits of integration then gives $\int_2^{10} 3x\sqrt{x-1} \, dx = \int_1^9 3(u+1)\sqrt{u} \, du$.

82. If $y' = \frac{2x-1}{\sqrt{2x^2-2x+3}}$, then $y =$

(A) $\sqrt{2x^2 - 2x + 3} + C$

(B) $2\sqrt{2x^2 - 2x + 3} + C$

(C) $\frac{1}{2}\sqrt{2x^2 - 2x + 3} + C$

(D) $\frac{x^2-x}{\sqrt{\frac{2x^3}{3}-x^2+3x}} + C$

Answer A

Correct. The antiderivative can be found by integrating $\frac{2x-1}{\sqrt{2x^2-2x+3}}$ using the method of substitution. Starting with the substitution $u = 2x^2 - 2x + 3$, $u = 2x^2 - 2x + 3 \Rightarrow \frac{du}{dx} = 4x - 2 = 2(2x - 1) \Rightarrow dx = \frac{du}{2(2x-1)}$.

Substituting for $2x^2 - 2x + 3$ and for dx gives $y = \int \frac{2x-1}{\sqrt{2x^2-2x+3}} dx = \int \frac{1}{\sqrt{u}} \cdot \frac{1}{2} du = \sqrt{u} + C = \sqrt{2x^2 - 2x + 3} + C$.

6-8-9

83. $\int_0^4 100e^{\frac{x}{4}} dx =$
- (A) $25(e - 1)$
- (B) $100(e - 1)$
- (C) $400(e - 1)$
- (D) $400(e^4 - 1)$



Answer C

Correct. Starting with the substitution $u = \frac{x}{4}$, $u = \frac{x}{4} \Rightarrow \frac{du}{dx} = \frac{1}{4} \Rightarrow dx = 4 du$.

When $x = 0$, $u = 0$.

When $x = 4$, $u = 1$.

Substituting for $\frac{x}{4}$, for dx , and for the limits, $\int_0^4 100e^{\frac{x}{4}} dx = \int_0^1 100e^u (4 du) = 400 \int_0^1 e^u du = 400e^u \Big|_{u=0}^{u=1} = 400e - 400 = 400(e - 1)$.

Alternatively, the antiderivative of $e^{\frac{x}{4}}$ is $e^{\frac{x}{4}}$ with an adjustment of dividing by $\frac{1}{4}$ to account for the chain rule. Therefore,

$$\int_0^4 100e^{\frac{x}{4}} dx = 100 \cdot \frac{e^{\frac{x}{4}}}{\frac{1}{4}} \Big|_{x=0}^{x=4} = 400e^{\frac{x}{4}} \Big|_{x=0}^{x=4} = 400e - 400 = 400(e - 1).$$

84. $\int \frac{1}{\sqrt{9 - x^2}} dx =$
- (A) $\ln(\sqrt{9 - x^2}) + C$
- (B) $\frac{1}{3}\sin^{-1}\left(\frac{x}{3}\right) + C$
- (C) $3\sin^{-1}\left(\frac{x}{3}\right) + C$
- (D) $\sin^{-1}\left(\frac{x}{3}\right) + C$



Answer D

Correct. This integral can be found using substitution of variables. The integrand should be recognized as coming from the derivative of an inverse sine function.

Let $u = \frac{x}{3}$, since that is the term being squared in the denominator after removing a factor of 9. Then, $du = \frac{dx}{3} \Rightarrow 3 du = dx$. The integral becomes the following.

$$\int \frac{1}{\sqrt{9 - x^2}} dx = 3 \int \frac{1}{\sqrt{9 - 9u^2}} du = 3 \int \frac{1}{3\sqrt{1 - u^2}} du = \sin^{-1} u + C = \sin^{-1}\left(\frac{x}{3}\right) + C$$

6-8-9

85. $\int_0^1 (x^3 + x)(x^4 + 2x^2 + 9)^{\frac{1}{2}} dx =$

(A) $\frac{1}{6}$

(B) $\frac{2}{3}$

(C) $4\sqrt{3} - \frac{9}{2}$

(D) $16\sqrt{3} - 18$



Answer C

Correct. Starting with the substitution $u = x^4 + 2x^2 + 9$, $u = x^4 + 2x^2 + 9 \Rightarrow du = (4x^3 + 4x) dx \Rightarrow \frac{1}{4} du = (x^3 + x) dx$.

Also, the limits of integration for x are changed to limits of integration for u .

$$x = 0 \Rightarrow u = 0 + 0 + 9 = 9$$

$$x = 1 \Rightarrow u = 1 + 2 + 9 = 12$$

Substituting for $x^4 + 2x^2 + 9$, for dx , and for the limits of integration gives the following.

$$\frac{1}{4} \int_9^{12} u^{\frac{1}{2}} du = \frac{1}{4} \cdot \frac{2}{3} u^{\frac{3}{2}} \Big|_9^{12} = \frac{1}{6} \left(12^{\frac{3}{2}} - 9^{\frac{3}{2}} \right) = \frac{24\sqrt{3}}{6} - \frac{27}{6} = 4\sqrt{3} - \frac{9}{2}$$

86. $\int \frac{3x^2 + 4x + 1}{3x^3 + 6x^2 + 3x + 5} dx =$

(A) $-\frac{9x^2 + 12x + 3}{(3x^3 + 6x^2 + 3x + 5)^2} + C$

(B) $\frac{x^3 + 2x^2 + x}{\frac{3}{4}x^4 + 2x^3 + \frac{3}{2}x^2 + 5x} + C$

(C) $\frac{1}{3} \ln |3x^3 + 6x^2 + 3x + 5| + C$

(D) $\ln |3x^3 + 6x^2 + 3x + 5| + C$



Answer C

Correct. The substitution $u = 3x^3 + 6x^2 + 3x + 5$ is used.

$$du = (9x^2 + 12x + 3) dx = 3(3x^2 + 4x + 1) dx \Rightarrow \frac{1}{3} du = (3x^2 + 4x + 1) dx$$

$$\int \frac{3x^2 + 4x + 1}{3x^3 + 6x^2 + 3x + 5} dx = \frac{1}{3} \int \frac{du}{u} = \frac{1}{3} \ln |u| + C = \frac{1}{3} \ln |3x^3 + 6x^2 + 3x + 5| + C$$

87. $\int \sin x \cos x e^{\sin^2 x + \pi} dx =$

(A) $-\sin x \cos x e^{\sin^2 x + \pi} + C$

(B) $e^{\sin^2 x + \pi} + C$

(C) $\frac{1}{2} \left(e^{\sin^2 x + \pi} \right) + C$

(D) $-\frac{1}{2} \left(e^{\sin^2 x + \pi} \right) + C$



Answer C

Correct. The substitution $u = \sin^2 x + \pi$ is used. $u = \sin^2 x + \pi \Rightarrow du = 2 \sin x \cos x dx \Rightarrow \frac{1}{2} du = \sin x \cos x dx$

$$\int \sin x \cos x e^{\sin^2 x + \pi} dx = \frac{1}{2} \int e^u du = \frac{1}{2} e^u + C = \frac{1}{2} \left(e^{\sin^2 x + \pi} \right) + C$$

6-8-9

88. Using the substitution $u = 5x - 2$, $\int x\sqrt{5x-2} \, dx$ is equivalent to which of the following?

(A) $\int u^{\frac{1}{2}} \, du$

(B) $\frac{1}{5} \int u^{\frac{1}{2}} \, du$

(C) $\frac{1}{5} \int \left(u^{\frac{3}{2}} + 2u^{\frac{1}{2}}\right) \, du$

(D) $\frac{1}{25} \int \left(u^{\frac{3}{2}} + 2u^{\frac{1}{2}}\right) \, du$



Answer D

Correct. The substitution $u = 5x - 2$ is used. $u = 5x - 2 \Rightarrow du = 5dx \Rightarrow \frac{1}{5}du = dx$

and $u = 5x - 2 \Rightarrow \frac{u+2}{5} = x$

Substituting for $5x - 2$, for x , and for dx gives the following.

$$\int x\sqrt{5x-2} \, dx = \int \frac{u+2}{5} \cdot \sqrt{u} \cdot \frac{1}{5} \, du = \frac{1}{25} \int (u+2)u^{\frac{1}{2}} \, du = \frac{1}{25} \int \left(u^{\frac{3}{2}} + 2u^{\frac{1}{2}}\right) \, du$$

- 89.



x	0	1	2	3
$g(x)$	5	4	1	-2

Selected values of the twice-differentiable function g are given in the table above. What is the value of $\int_0^3 g'(x)\cos^2(2g(x)+1) \, dx$?

(A) -13.856

(B) -3.464

(C) -1.587

(D) 0.715



Answer B

Correct. Use the technique of substitution of variables with $u = 2g(x) + 1$.

$$u = 2g(x) + 1 \Rightarrow du = 2g'(x)dx \Rightarrow g'(x)dx = \frac{1}{2}du$$

$$\text{When } x = 0, u = 2g(0) + 1 = 2(5) + 1 = 11.$$

$$\text{When } x = 3, u = 2g(3) + 1 = 2(-2) + 1 = -3.$$

$$\int_0^3 g'(x)\cos^2(2g(x)+1) \, dx = \frac{1}{2} \int_{11}^{-3} \cos^2 u \, du = -3.464$$

The calculator is used to do the numerical integration of the definite integral.

6-8-9

90. Using the substitution $u = 4x - 3$, $\int x(4x - 3)^{10} dx$ is equivalent to which of the following?

(A) $\int u^{10} du$

(B) $\frac{1}{4} \int u^{10} du$

(C) $\frac{1}{4} \int (u^{11} + 3u^{10}) du$

(D) $\frac{1}{16} \int (u^{11} + 3u^{10}) du$

**Answer D**

Correct. The substitution $u = 4x - 3$ is used.

$$u = 4x - 3 \Rightarrow du = 4dx \Rightarrow dx = \frac{1}{4}du$$

$$u = 4x - 3 \Rightarrow x = \frac{1}{4}(u + 3)$$

Substituting for $4x - 3$, for x , and for dx gives the following.

$$\int x(4x - 3)^{10} dx = \int \frac{1}{4}(u + 3) \cdot u^{10} \cdot \frac{1}{4} du = \frac{1}{16} \int (u^{11} + 3u^{10}) du$$

91. $\int e^x \sin(6e^x + 3) dx =$

(A) $-e^x \cos(6e^x + 3) + C$

(B) $-6 \cos(6e^x + 3) + C$

(C) $-\frac{1}{6} \cos(6e^x + 3) + C$

(D) $\frac{1}{6} \cos(6e^x + 3) + C$

**Answer C**

Correct. The substitution $u = 6e^x + 3$ is used.

$$u = 6e^x + 3 \Rightarrow du = 6e^x dx \Rightarrow dx = \frac{1}{6e^x} du$$

$$\int e^x \sin(6e^x + 3) dx = \frac{1}{6} \int \sin u du = -\frac{1}{6} \cos u + C = -\frac{1}{6} \cos(6e^x + 3) + C$$

6-8-9

92. $\int \frac{4x^4 + 3}{4x^5 + 15x + 2} dx =$

(A) $-\frac{1}{5(4x^5+15x+2)^2} + C$

(B) $\frac{\frac{4}{3}x^5+3x}{\frac{2}{3}x^6+\frac{15}{2}x^2+2x} + C$

(C) $\frac{1}{5} \ln |4x^5 + 15x + 2| + C$

(D) $5 \ln |4x^5 + 15x + 2| + C$

Answer C

Correct. The substitution $u = 4x^5 + 15x + 2$ is used.

$$u = 4x^5 + 15x + 2 \Rightarrow du = (20x^4 + 15)dx = 5(4x^4 + 3)dx \Rightarrow dx = \frac{1}{5(4x^4+3)} du$$

$$\int \frac{4x^4 + 3}{4x^5 + 15x + 2} dx = \frac{1}{5} \int \frac{1}{u} du = \frac{1}{5} \ln |u| + C = \frac{1}{5} \ln |4x^5 + 15x + 2| + C$$

Note: The substitution $u = 4x^5 + 15x$ could also be used to get $\frac{1}{5} \int \frac{1}{u+2} du = \frac{1}{5} \ln |u+2| + C$.

93. $\int_0^1 (x+2)(3x^2+12x+1)^{\frac{1}{2}} dx =$

(A) $\frac{1}{9}$

(B) 4

(C) 7

(D) 252

Answer C

Correct. Starting with the substitution $u = 3x^2 + 12x + 1$, $u = 3x^2 + 12x + 1 \Rightarrow du = (6x + 12)dx \Rightarrow dx = \frac{1}{6} \cdot \frac{1}{(x+2)} du$.

Also, the limits of integration for x are changed to limits of integration for u .

$$x = 0 \Rightarrow u = 0 + 0 + 1 = 1$$

$$x = 1 \Rightarrow u = 3 + 12 + 1 = 16$$

Substituting for $3x^2 + 12x + 1$, for dx , and for the limits of integration gives the following.

$$\frac{1}{6} \int_1^{16} u^{\frac{1}{2}} du = \frac{1}{6} \cdot \frac{2}{3} u^{\frac{3}{2}} \Big|_1^{16} = \frac{1}{9} (16)^{\frac{3}{2}} - \frac{1}{9} (1) = \frac{64}{9} - \frac{1}{9} = \frac{63}{9} = 7$$

94. If $f'(x) = 6x^2 - 2x + 5$ and $f(1) = -3$, then $f(2) =$

(A) -19

(B) 8

(C) 13

(D) 22

Answer C

Correct. Using the Fundamental Theorem of Calculus gives the following.

$$f(2) = f(1) + \int_1^2 f'(x) dx = f(1) + \int_1^2 (6x^2 - 2x + 5) dx = -3 + (2x^3 - x^2 + 5x) \Big|_1^2 = -3 + (16 - 4 + 10) - (2 - 1 + 5) = -3 + 22 - 6 = 13$$

An alternative approach is to first solve for $f(x)$ to get $f(x) = 2x^3 - x^2 + 5x + C$. Using the initial condition,

$$-3 = f(1) = 2 - 1 + 5 + C = 6 + C \Rightarrow C = -9. \text{ Then } f(2) = 16 - 4 + 10 - 9 = 13.$$

6-8-9

95. Let R be the region between the graphs of $y = 1 + \sin(\pi x)$ and $y = x^2$ from $x = 0$ to $x = 1$.

(a) Find the area of R .

(b) Set up, but do not integrate an integral expression in terms of a single variable for the volume of the solid generated when R is revolved about the x -axis.

(c) Set up, but do not integrate an integral expression in terms of a single variable for the volume of the solid generated when R is revolved about the y -axis.

Part A

2 points are earned for integral

1 point is **deducted** for limit error

2 points are **deducted** for integrand error

2 points are **deducted** for reversed integral

3 points are earned for antiderivative and evaluate

3 points are **deducted** for constant term

2 points are **deducted** for chain rule error

1 point is **deducted** for each other error

$$\begin{aligned} A &= \int_0^1 1 + \sin \pi x - x^2 dx \\ &= \left(x - \frac{1}{\pi} \cos \pi x - \frac{1}{3} x^3 \right) \Big|_0^1 \\ &= \left(1 - \frac{1}{\pi}(-1) - \frac{1}{3} \right) - \left(0 - \frac{1}{\pi} - 0 \right) \\ &= \frac{2}{3} + \frac{2}{\pi} \end{aligned}$$



0	1	2	3	4	5
---	---	---	---	---	---

The student response earns all of the following points:

2 points are earned for integral

1 point is **deducted** for limit error

2 points are **deducted** for integrand error

2 points are **deducted** for reversed integral

3 points are earned for antiderivative and evaluate

3 points are **deducted** for constant term

2 points are **deducted** for chain rule error

1 point is **deducted** for each other error

6-8-9

$$\begin{aligned}
 A &= \int_0^1 1 + \sin \pi x - x^2 dx \\
 &= \left(x - \frac{1}{\pi} \cos \pi x - \frac{1}{3} x^3 \right) \Big|_0^1 \\
 &= \left(1 - \frac{1}{\pi}(-1) - \frac{1}{3} \right) - \left(0 - \frac{1}{\pi} - 0 \right) \\
 &= \frac{2}{3} + \frac{2}{\pi}
 \end{aligned}$$

Part B

2 points are earned for integral expression

1 point is **deducted** for incorrect limits

$$V = \pi \int_0^1 (1 + \sin \pi x)^2 - x^4 dx$$

or

$$2\pi \int_0^1 y^{3/2} dy + 2\pi \int_1^2 y \left(1 - \frac{2}{\pi} \arcsin(y-1) \right) dy$$

✓

0	1	2
---	---	---

The student response earns all of the following points:

2 points are earned for integral expression

1 point is **deducted** for incorrect limits

$$V = \pi \int_0^1 (1 + \sin \pi x)^2 - x^4 dx$$

or

$$2\pi \int_0^1 y^{3/2} dy + 2\pi \int_1^2 y \left(1 - \frac{2}{\pi} \arcsin(y-1) \right) dy$$

Part C

2 points are earned for integral

1 point is **deducted** for π or 2π error

2 points are **deducted** for integrand error (including reverse)

1 point is **deducted** for one or more limit error

6-8-9

$$V = 2\pi \int_0^1 x(1 + \sin(\pi x) - x^2) dx$$

or

$$\pi \int_0^1 y dy + \pi \int_1^2 \left(1 - \frac{1}{\pi} \arcsin(y-1)\right)^2 - \left(\frac{1}{\pi} \arcsin(y-1)\right)^2 dy$$

✓

0	1	2
---	---	---

The student response earns all of the following points:

2 points are earned for integral

1 point is **deducted** for π or 2π error

2 points are **deducted** for integrand error (including reverse)

1 point is **deducted** for one or more limit error

$$V = 2\pi \int_0^1 x(1 + \sin(\pi x) - x^2) dx$$

or

$$\pi \int_0^1 y dy + \pi \int_1^2 \left(1 - \frac{1}{\pi} \arcsin(y-1)\right)^2 - \left(\frac{1}{\pi} \arcsin(y-1)\right)^2 dy$$

96. The function f is given by $f(x) = 4x^3 - x^4$. On what intervals is the graph of f concave up?

(A) $(-\infty, 0)$ and $(2, \infty)$

(B) $(-\infty, 3)$

(C) $(0, 2)$ only

(D) $(0, 3)$ only

✓

Answer C

Correct. The graph of f will be concave up on intervals where $f''(x) > 0$.

$$f'(x) = 12x^2 - 4x^3$$

$$f''(x) = 24x - 12x^2 = 12x(2 - x)$$

The graph of f'' is a parabola opening downward and with zeros at $x = 0$ and $x = 2$. Therefore, $f''(x) > 0$ on the interval between the two zeros, or $0 < x < 2$.

6-8-9

97.

x	-2	$-2 < x < -1$	-1	$-1 < x < 1$	1	$1 < x < 3$	3
$f(x)$	12	Positive	8	Positive	2	Positive	7
$f'(x)$	-5	Negative	0	Negative	0	Positive	$\frac{1}{2}$
$g(x)$	-1	Negative	0	Positive	3	Positive	1
$g'(x)$	2	Positive	$\frac{3}{2}$	Positive	0	Negative	-2

The twice-differentiable functions f and g are defined for all real numbers x . Values of f , f' , g , and g' for various values of x are given in the table above.

- (a) Find the x -coordinate of each relative minimum of f on the interval $[-2, 3]$. Justify your answers.
- (b) Explain why there must be a value c , for $-1 < c < 1$, such that $f''(c) = 0$.
- (c) The function h is defined by $h(x) = \ln(f(x))$. Find $h'(3)$. Show the computations that lead to your answer.
- (d) Evaluate $\int_{-2}^3 f'(g(x))g'(x) dx$.

Part A

1 point is earned for the answer with justification

$x = 1$ is the only critical point at which f' changes sign from negative to positive. Therefore, f has a relative minimum at $x = 1$.



0	1
---	---

The student response earns all of the following points:

1 point is earned for the answer with justification

$x = 1$ is the only critical point at which f' changes sign from negative to positive. Therefore, f has a relative minimum at $x = 1$.

Part B

1 point is earned for $f'(1) - f'(-1) = 0$

1 point is earned for the explanation, using Mean Value Theorem

f' is differentiable $\Rightarrow f'$ is continuous on the interval $-1 \leq x \leq 1$

$$\frac{f'(1) - f'(-1)}{1 - (-1)} = \frac{0 - 0}{2} = 0$$

Therefore, by the Mean Value Theorem, there is at least one value c , $-1 < c < 1$, such that $f''(c) = 0$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $f'(1) - f'(-1) = 0$

1 point is earned for the explanation, using Mean Value Theorem

f' is differentiable $\Rightarrow f'$ is continuous on the interval $-1 \leq x \leq 1$

$$\frac{f'(1) - f'(-1)}{1 - (-1)} = \frac{0 - 0}{2} = 0$$

6-8-9

Therefore, by the Mean Value Theorem, there is at least one value c , $-1 < c < 1$, such that $f'(c) = 0$.

Part C

2 points are earned for $h'(x)$

1 point is earned for the answer

$$h'(x) = \frac{1}{f(x)} \cdot f'(x)$$

$$h'(3) = \frac{1}{f(3)} \cdot f'(3) = \frac{1}{7} \cdot \frac{1}{2} = \frac{1}{14}$$

✓

0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for $h'(x)$

1 point is earned for the answer

$$h'(x) = \frac{1}{f(x)} \cdot f'(x)$$

$$h'(3) = \frac{1}{f(3)} \cdot f'(3) = \frac{1}{7} \cdot \frac{1}{2} = \frac{1}{14}$$

Part D

2 points are earned for stating Fundamental Theorem of Calculus

1 point is earned for the answer

$$\begin{aligned} \int_{-2}^3 f'(g(x))g'(x) dx &= [f(g(x))]_{x=-2}^{x=3} \\ &= f(g(3)) - f(g(-2)) \\ &= f(1) - f(-1) \\ &= 2 - 8 = -6 \end{aligned}$$

✓

0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for stating Fundamental Theorem of Calculus

1 point is earned for the answer

$$\begin{aligned} \int_{-2}^3 f'(g(x))g'(x) dx &= [f(g(x))]_{x=-2}^{x=3} \\ &= f(g(3)) - f(g(-2)) \\ &= f(1) - f(-1) \\ &= 2 - 8 = -6 \end{aligned}$$